

Indefinite Integration

Worked Solutions with TikZ Graphs

Wisest Maths — Year 1 A-Level, Integration (ref i1)

This document contains all 70 *Indefinite Integration* questions, each with a fully worked solution and a TikZ graph. Because an indefinite integral is only determined up to a constant, each diagram shows the *family of antiderivative curves* $y = F(x) + C$ (grey, vertical translates of one another), with one representative member drawn in blue. For the questions with a boundary condition, the particular curve is highlighted and the given point is marked — showing how the condition fixes the value of C .

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Indefinite Integration 01 (i1-001)

Difficulty: Foundation / *Marks: 1*

Find $\int x^6 \, dx$.

Worked solution.

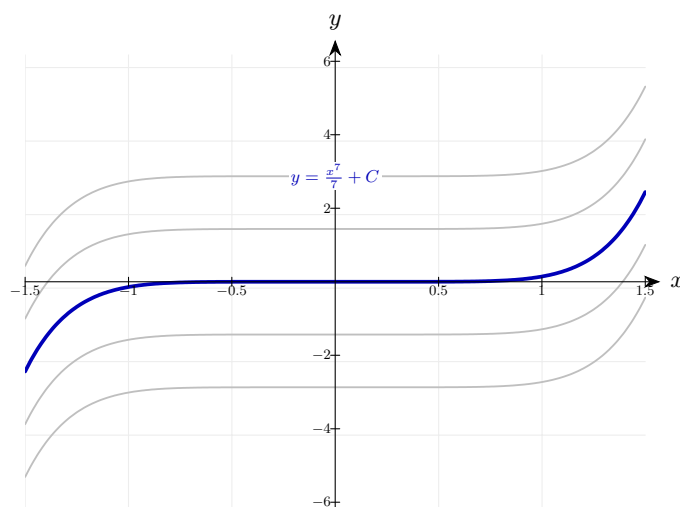
Apply the integration power rule: increase the index by 1 and divide by the new index.

$$\int x^6 \, dx = \frac{x^7}{7} + C$$

The rule is $\int x^n \, dx = \frac{x^{n+1}}{n+1} + C$. Here $n = 6$, so the index becomes 7.

Final answer: $\frac{x^7}{7} + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 02 (i1-002)

Difficulty: Foundation / Marks: 1

Find $\int 5x^4 dx$.

Worked solution.

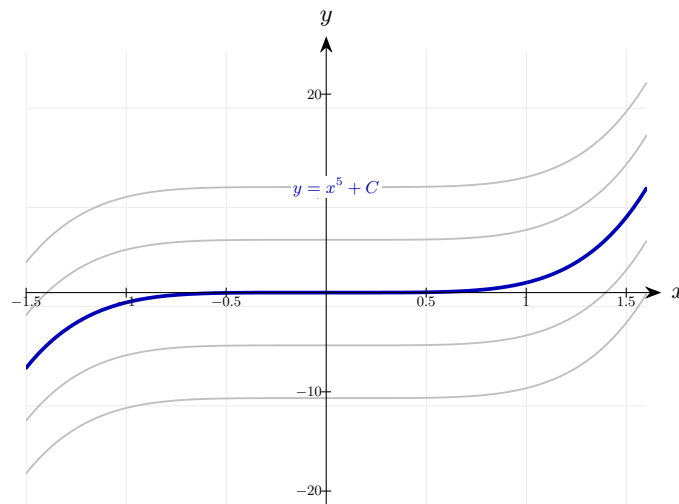
Take the constant outside the integral, then integrate.

$$\int 5x^4 dx = 5 \cdot \frac{x^5}{5} + C = x^5 + C$$

The constant 5 multiplies the integral. Increase the index from 4 to 5 and divide by 5; the 5s cancel.

Final answer: $x^5 + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 03 (i1-003)

Difficulty: Foundation / Marks: 1

Find $\int 8x dx$.

Worked solution.

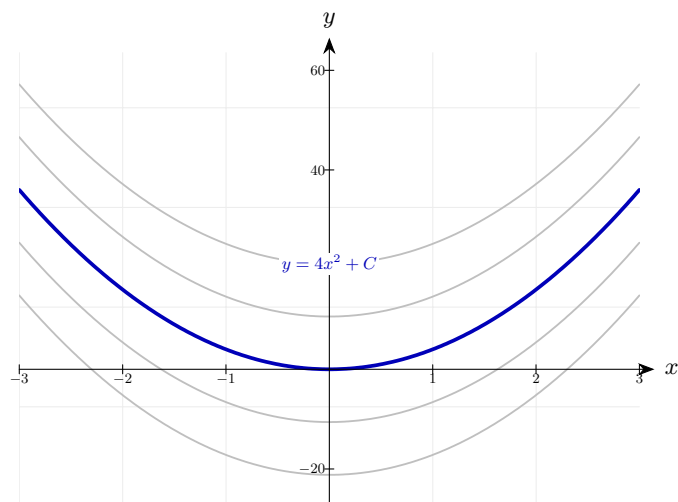
Write $8x = 8x^1$ and apply the power rule.

$$\int 8x dx = 8 \cdot \frac{x^2}{2} + C = 4x^2 + C$$

Increase the index from 1 to 2 and divide by 2; $8 \div 2 = 4$.

Final answer: $4x^2 + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 04 (i1-004)

Difficulty: Foundation / Marks: 1

Find $\int 12 \, dx$.

Worked solution.

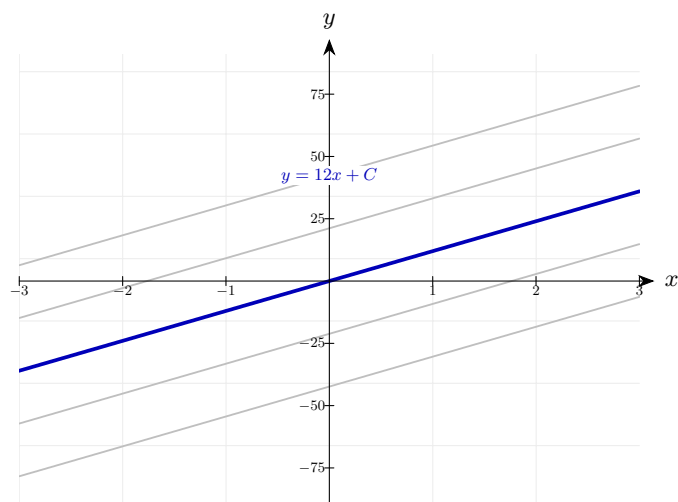
Write 12 as $12x^0$ and integrate.

$$\int 12 \, dx = 12x + C$$

Integrating a constant k gives $kx + C$. The index increases from 0 to 1 and we divide by 1.

Final answer: $12x + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 05 (i1-005)

Difficulty: Foundation / Marks: 1

Find $\int -3x^2 \, dx$.

Worked solution.

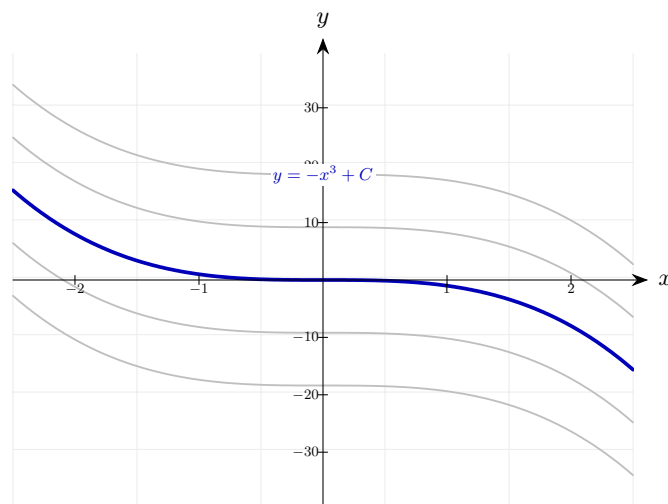
Apply the power rule with the negative constant.

$$\int -3x^2 \, dx = -3 \cdot \frac{x^3}{3} + C = -x^3 + C$$

Increase the index from 2 to 3 and divide by 3; $-3 \div 3 = -1$.

Final answer: $-x^3 + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 06 (i1-006)

Difficulty: Foundation / Marks: 2

Find $\int x^{-3} \, dx$.

Worked solution.

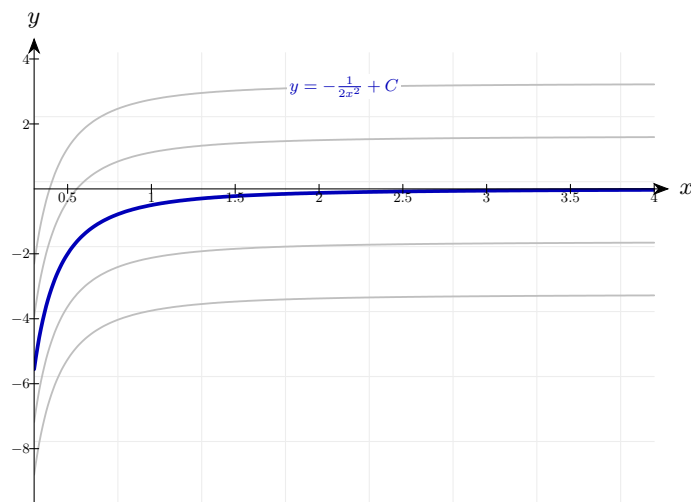
Apply the power rule with $n = -3$.

$$\int x^{-3} \, dx = \frac{x^{-2}}{-2} + C = -\frac{1}{2x^2} + C$$

Increase the index from -3 to -2 and divide by -2 .

Final answer: $-\frac{1}{2x^2} + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 07 (i1-007)

Difficulty: Foundation / Marks: 2

Find $\int \sqrt{x} \, dx$.

Worked solution.

Step 1: Rewrite $\sqrt{x} = x^{1/2}$.

$$\int x^{1/2} \, dx$$

Always convert surds to fractional powers before integrating.

Step 2: Apply the power rule.

$$= \frac{x^{3/2}}{3/2} + C = \frac{2}{3}x^{3/2} + C$$

Increase the index from $\frac{1}{2}$ to $\frac{3}{2}$ and divide by $\frac{3}{2}$ — dividing by a fraction is the same as multiplying by its reciprocal.

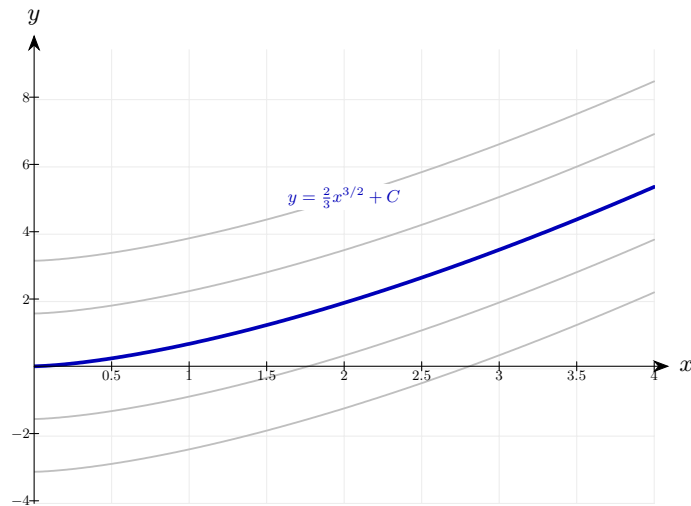
Step 3: Rewrite using surd notation if required.

$$= \frac{2}{3}x\sqrt{x} + C$$

$$x^{3/2} = x \cdot \sqrt{x}.$$

Final answer: $\frac{2}{3}x^{3/2} + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 08 (i1-008)

Difficulty: Foundation / Marks: 2

Find $\int \frac{4}{x^2} dx$.

Worked solution.

Step 1: Rewrite as a negative power of x .

$$\int 4x^{-2} dx$$

$$\frac{4}{x^2} = 4x^{-2}.$$

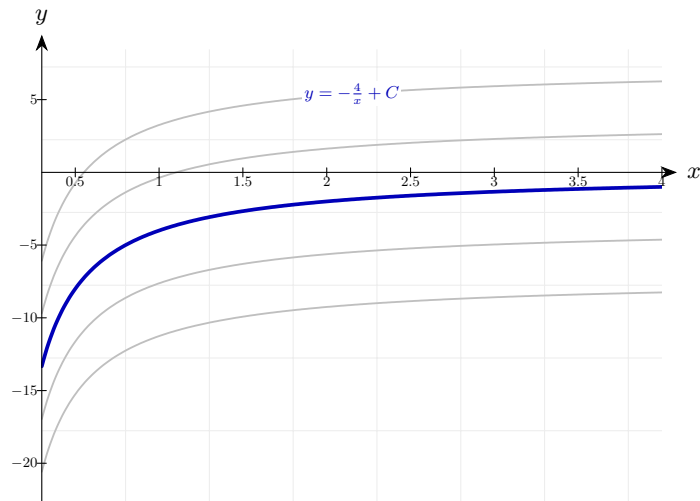
Step 2: Integrate.

$$= 4 \cdot \frac{x^{-1}}{-1} + C = -4x^{-1} + C = -\frac{4}{x} + C$$

Increase the index from -2 to -1 and divide by -1 .

Final answer: $-\frac{4}{x} + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 09 (i1-009)

Difficulty: Foundation / Marks: 3

Find $\int (3x^2 + 5x - 2) \, dx$.

Worked solution.

Step 1: Integrate each term separately.

$$\int 3x^2 \, dx + \int 5x \, dx + \int (-2) \, dx$$

The integral of a sum equals the sum of the integrals.

Step 2: Apply the power rule to each term.

$$= 3 \cdot \frac{x^3}{3} + 5 \cdot \frac{x^2}{2} - 2x + C$$

Increase each index by 1 and divide by the new index.

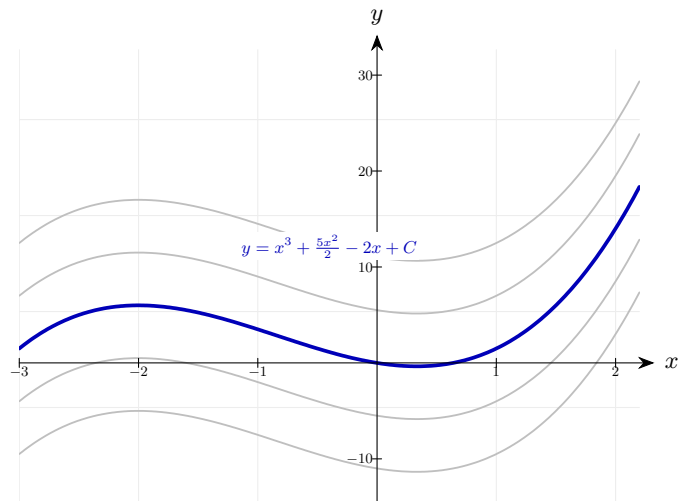
Step 3: Simplify.

$$= x^3 + \frac{5x^2}{2} - 2x + C$$

$3 \div 3 = 1$. Only one constant of integration is needed for the whole expression.

Final answer: $x^3 + \frac{5x^2}{2} - 2x + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 10 (i1-010)

Difficulty: Foundation / Marks: 3

Find $\int (4x^3 - 6x^2 + 1) dx$.

Worked solution.

Step 1: Integrate each term separately.

$$= 4 \cdot \frac{x^4}{4} - 6 \cdot \frac{x^3}{3} + x + C$$

Apply $\int x^n dx = \frac{x^{n+1}}{n+1}$ to each term.

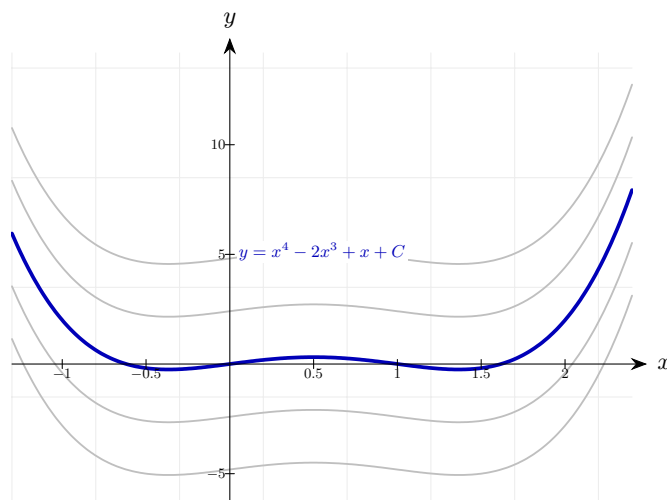
Step 2: Simplify.

$$= x^4 - 2x^3 + x + C$$

$4 \div 4 = 1$ and $6 \div 3 = 2$.

Final answer: $x^4 - 2x^3 + x + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 11 (i1-011)

Difficulty: Foundation / Marks: 3

Expand and then integrate $(x + 3)(x - 2)$.

Worked solution.

Step 1: Expand the bracket.

$$(x + 3)(x - 2) = x^2 + x - 6$$

FOIL: $x^2 - 2x + 3x - 6$.

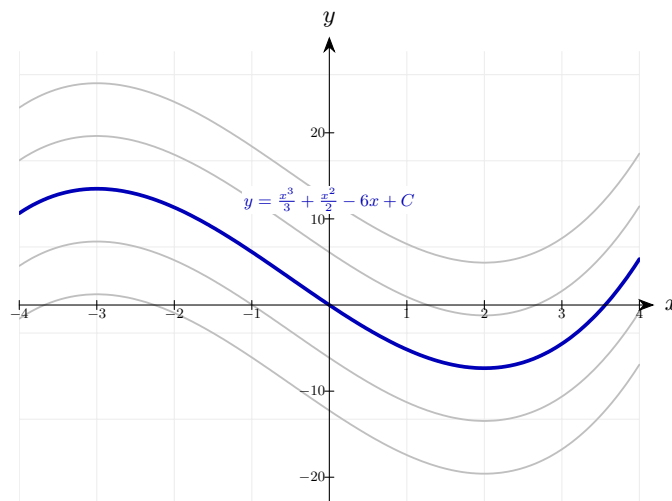
Step 2: Integrate term by term.

$$\int (x^2 + x - 6) dx = \frac{x^3}{3} + \frac{x^2}{2} - 6x + C$$

Apply the power rule to each term.

Final answer: $\frac{x^3}{3} + \frac{x^2}{2} - 6x + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 12 (i1-012)

Difficulty: Foundation / Marks: 3

Expand and integrate $x^2(3x - 5)$.

Worked solution.

Step 1: Expand.

$$x^2(3x - 5) = 3x^3 - 5x^2$$

Multiply x^2 by each term inside the bracket.

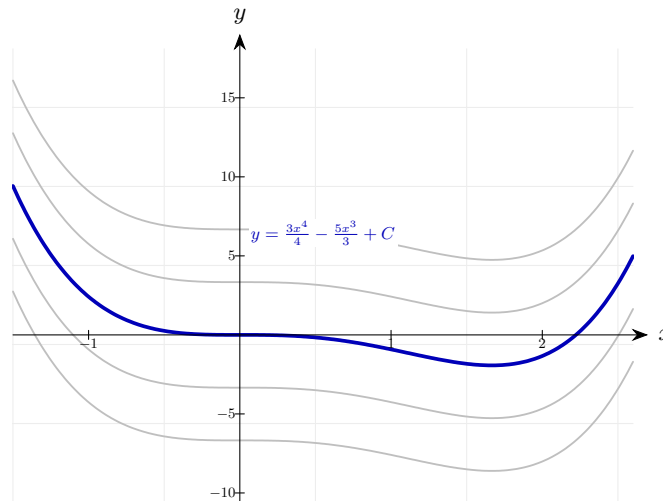
Step 2: Integrate.

$$\int (3x^3 - 5x^2) dx = \frac{3x^4}{4} - \frac{5x^3}{3} + C$$

Apply the power rule to each term.

Final answer: $\frac{3x^4}{4} - \frac{5x^3}{3} + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 13 (i1-013)

Difficulty: Foundation / Marks: 3

Find $\int (2x - 1)^2 dx$.

Worked solution.

Step 1: Expand $(2x - 1)^2$.

$$(2x - 1)^2 = 4x^2 - 4x + 1$$

$(a - b)^2 = a^2 - 2ab + b^2$ with $a = 2x$, $b = 1$.

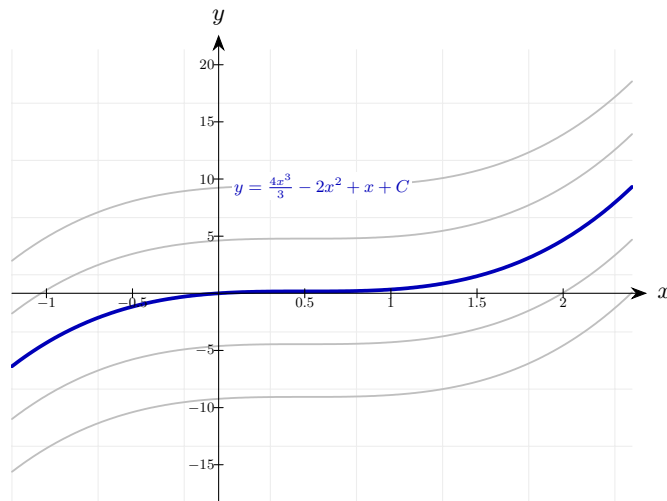
Step 2: Integrate each term.

$$\int (4x^2 - 4x + 1) dx = \frac{4x^3}{3} - 2x^2 + x + C$$

Power rule applied to each term.

Final answer: $\frac{4x^3}{3} - 2x^2 + x + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 14 (i1-014)

Difficulty: Foundation / Marks: 3

Find $\int (x^{1/3} + 2x) \, dx$.

Worked solution.

Step 1: Integrate each term.

$$\int x^{1/3} \, dx + \int 2x \, dx$$

Integrate term by term.

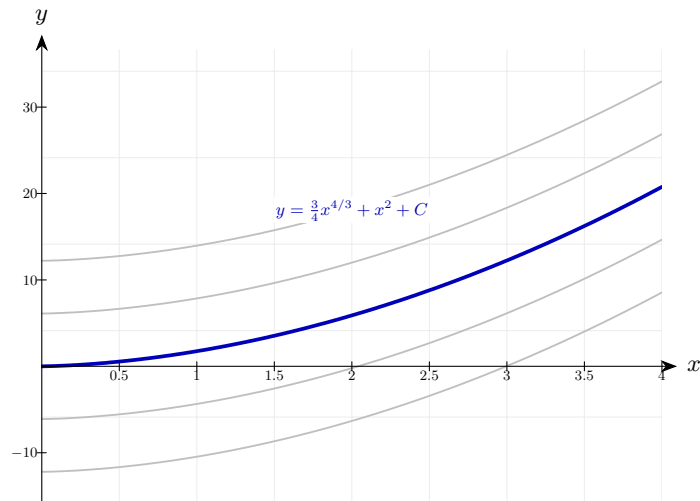
Step 2: Apply the power rule.

$$= \frac{x^{4/3}}{4/3} + x^2 + C = \frac{3}{4}x^{4/3} + x^2 + C$$

For $x^{1/3}$: new index = $\frac{4}{3}$; dividing by $\frac{4}{3}$ gives $\times \frac{3}{4}$. For $2x$: $2 \cdot \frac{x^2}{2} = x^2$.

Final answer: $\frac{3}{4}x^{4/3} + x^2 + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 15 (i1-015)

Difficulty: Foundation / Marks: 3

Find $\int \frac{x^3 - 4x}{x} dx$.

Worked solution.

Step 1: Simplify the fraction by dividing each term by x .

$$\frac{x^3 - 4x}{x} = x^2 - 4$$

$$\frac{x^3}{x} = x^2 \text{ and } \frac{-4x}{x} = -4.$$

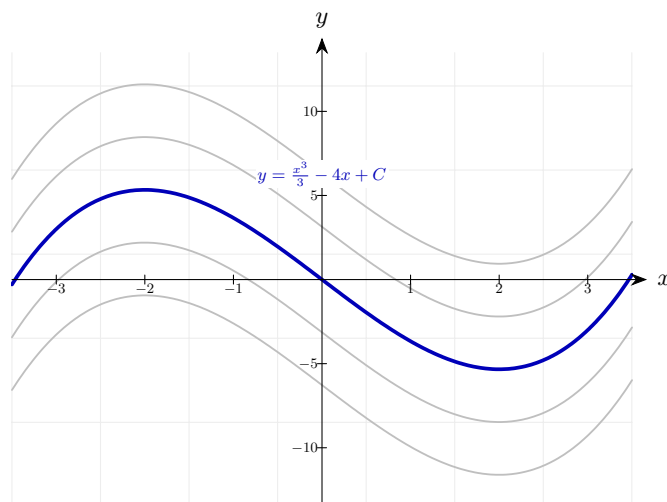
Step 2: Integrate.

$$\int (x^2 - 4) dx = \frac{x^3}{3} - 4x + C$$

Power rule applied to x^2 ; integrating the constant -4 gives $-4x$.

Final answer: $\frac{x^3}{3} - 4x + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 16 (i1-016)

Difficulty: Foundation / Marks: 3

Find $\int \frac{6x^4 + 3x^2}{3x} dx$.

Worked solution.

Step 1: Divide each term in the numerator by $3x$.

$$\frac{6x^4}{3x} + \frac{3x^2}{3x} = 2x^3 + x$$

$$\frac{6x^4}{3x} = 2x^3 \text{ and } \frac{3x^2}{3x} = x.$$

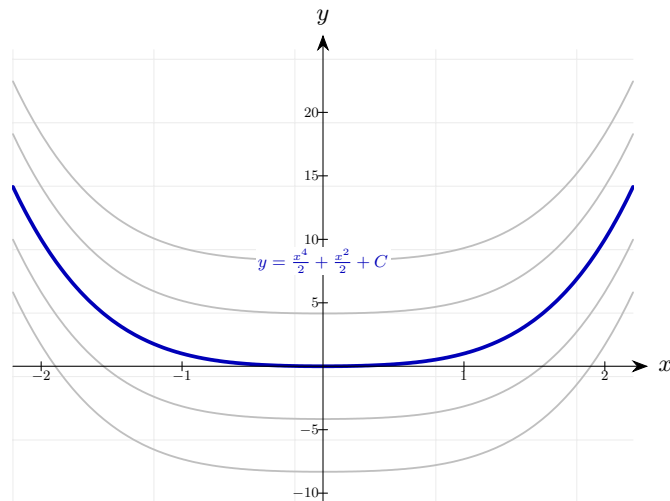
Step 2: Integrate.

$$\int (2x^3 + x) dx = \frac{x^4}{2} + \frac{x^2}{2} + C$$

$$2 \cdot \frac{x^4}{4} = \frac{x^4}{2} \text{ and } \frac{x^2}{2}.$$

Final answer: $\frac{x^4}{2} + \frac{x^2}{2} + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 17 (i1-017)

Difficulty: Foundation / Marks: 4

Find $\int \left(3\sqrt{x} - \frac{2}{x^2} \right) dx$.

Worked solution.

Step 1: Rewrite using index notation.

$$3x^{1/2} - 2x^{-2}$$

$$\sqrt{x} = x^{1/2} \text{ and } \frac{1}{x^2} = x^{-2}.$$

Step 2: Integrate each term.

$$\int 3x^{1/2} dx - \int 2x^{-2} dx = 3 \cdot \frac{x^{3/2}}{3/2} - 2 \cdot \frac{x^{-1}}{-1} + C$$

Apply the power rule to each term.

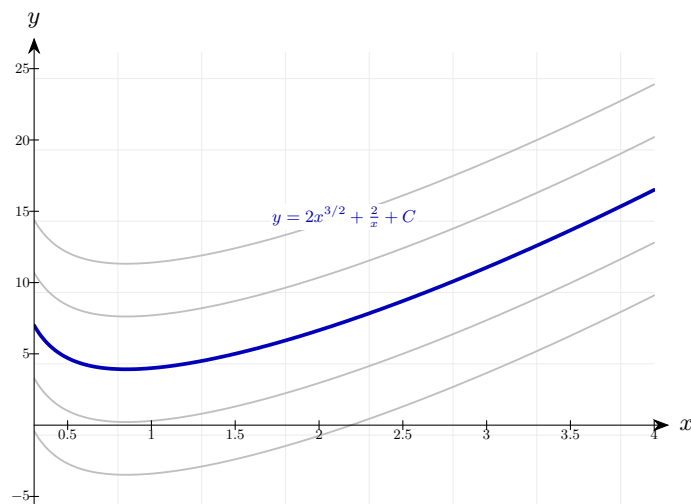
Step 3: Simplify.

$$= 2x^{3/2} + \frac{2}{x} + C$$

$$3 \div \frac{3}{2} = 3 \times \frac{2}{3} = 2. \quad -2 \div (-1) = 2.$$

Final answer: $2x^{3/2} + \frac{2}{x} + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 18 (i1-018)

Difficulty: Foundation / Marks: 4

Find $\int \sqrt{x}(x+4) \, dx$.

Worked solution.

Step 1: Expand by multiplying $x^{1/2}$ through the bracket.

$$\sqrt{x}(x+4) = x^{3/2} + 4x^{1/2}$$

$x^{1/2} \cdot x = x^{3/2}$ and $x^{1/2} \cdot 4 = 4x^{1/2}$.

Step 2: Integrate each term.

$$\int (x^{3/2} + 4x^{1/2}) \, dx = \frac{x^{5/2}}{5/2} + 4 \cdot \frac{x^{3/2}}{3/2} + C$$

Power rule applied to each fractional-index term.

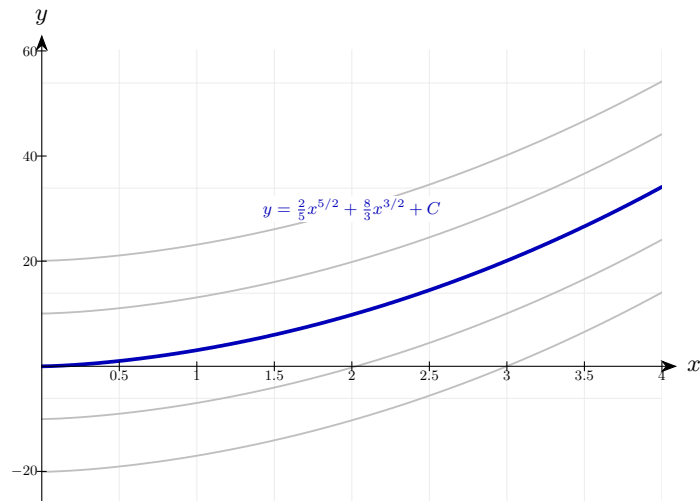
Step 3: Simplify.

$$= \frac{2}{5}x^{5/2} + \frac{8}{3}x^{3/2} + C$$

Dividing by $\frac{5}{2}$ gives $\times \frac{2}{5}$; dividing by $\frac{3}{2}$ gives $\times \frac{2}{3}$, then $\times 4 = \frac{8}{3}$.

Final answer: $\frac{2}{5}x^{5/2} + \frac{8}{3}x^{3/2} + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 19 (i1-019)

Difficulty: Foundation / Marks: 4

Find $\int \frac{x^2 + 5x + 6}{\sqrt{x}} dx$.

Worked solution.

Step 1: Divide each term by $x^{1/2}$.

$$x^{3/2} + 5x^{1/2} + 6x^{-1/2}$$

$$\frac{x^2}{x^{1/2}} = x^{3/2}, \frac{5x}{x^{1/2}} = 5x^{1/2}, \frac{6}{x^{1/2}} = 6x^{-1/2}.$$

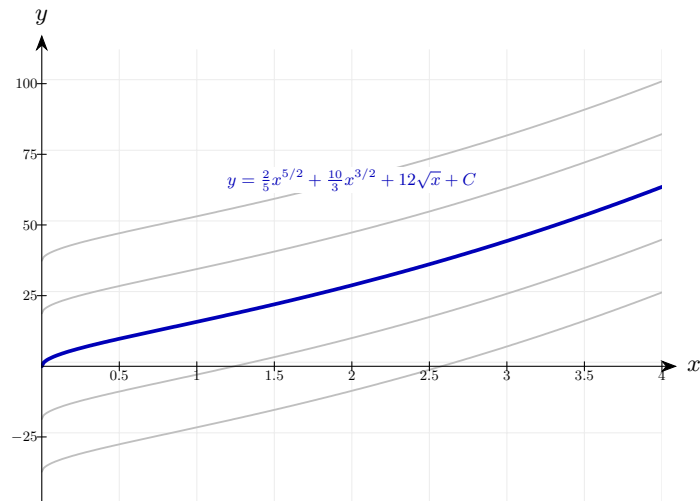
Step 2: Integrate each term.

$$\int (x^{3/2} + 5x^{1/2} + 6x^{-1/2}) dx = \frac{2}{5}x^{5/2} + \frac{10}{3}x^{3/2} + 12x^{1/2} + C$$

Apply $\int x^n dx = \frac{x^{n+1}}{n+1}$ to each term.

$$\textbf{Final answer: } \frac{2}{5}x^{5/2} + \frac{10}{3}x^{3/2} + 12\sqrt{x} + C$$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 20 (i1-020)

Difficulty: Foundation / Marks: 3

Find y given that $\frac{dy}{dx} = 6x^2 - 4x + 3$.

Worked solution.

Step 1: Integrate the right-hand side with respect to x .

$$y = \int (6x^2 - 4x + 3) \, dx$$

Integrating $\frac{dy}{dx}$ gives y .

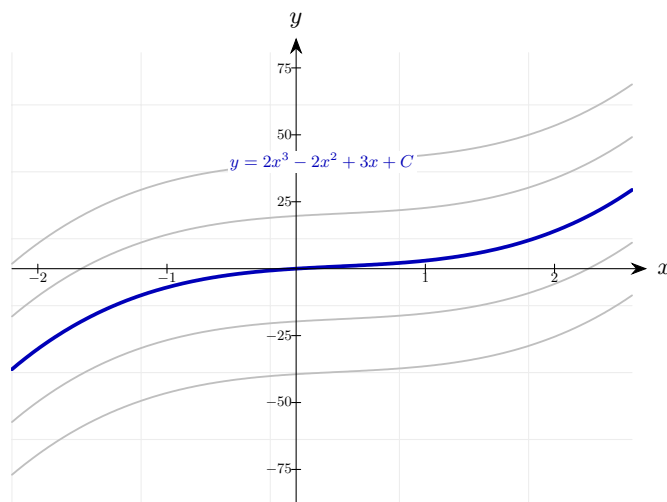
Step 2: Apply the power rule to each term.

$$y = 2x^3 - 2x^2 + 3x + C$$

$$6 \cdot \frac{x^3}{3} = 2x^3, 4 \cdot \frac{x^2}{2} = 2x^2, 3x^0 \rightarrow 3x.$$

Final answer: $y = 2x^3 - 2x^2 + 3x + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 21 (i1-021)

Difficulty: Foundation / Marks: 4

The curve $y = f(x)$ passes through the point $(0, 3)$ and $f'(x) = 4x - 1$. Find $f(x)$.

Worked solution.

Step 1: Integrate $f'(x)$.

$$f(x) = \int (4x - 1) dx = 2x^2 - x + C$$

Power rule applied to each term.

Step 2: Use the point $(0, 3)$ to find C .

$$f(0) = 3 : 0 - 0 + C = 3 \implies C = 3$$

Substitute $x = 0$ and $f(0) = 3$.

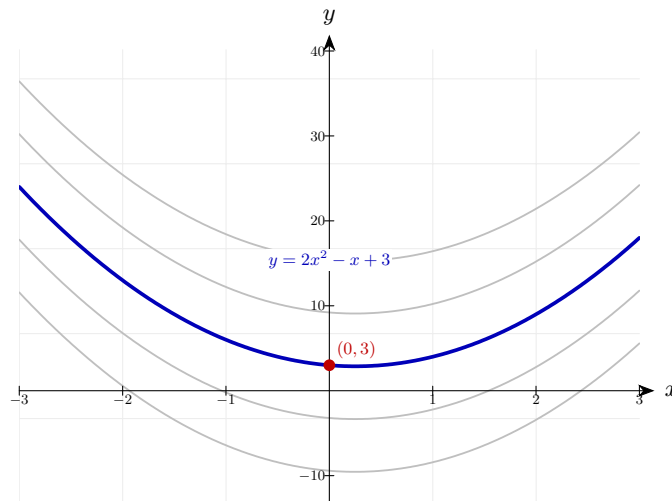
Step 3: Write the full equation.

$$f(x) = 2x^2 - x + 3$$

Substitute the found value of C .

Final answer: $f(x) = 2x^2 - x + 3$

Diagram.



The family of antiderivatives (grey) with the particular solution (blue) through the given point.

Indefinite Integration 22 (i1-022)

Difficulty: Foundation / Marks: 4

A curve passes through $(2, 5)$ and has gradient function $f'(x) = 3x^2 - 6x + 1$. Find $f(x)$.

Worked solution.

Step 1: Integrate the gradient function.

$$f(x) = \int (3x^2 - 6x + 1) dx = x^3 - 3x^2 + x + C$$

Apply the power rule to each term.

Step 2: Substitute $(2, 5)$ to find C .

$$5 = 8 - 12 + 2 + C \implies 5 = -2 + C \implies C = 7$$

Replace $x = 2$ and $f(x) = 5$.

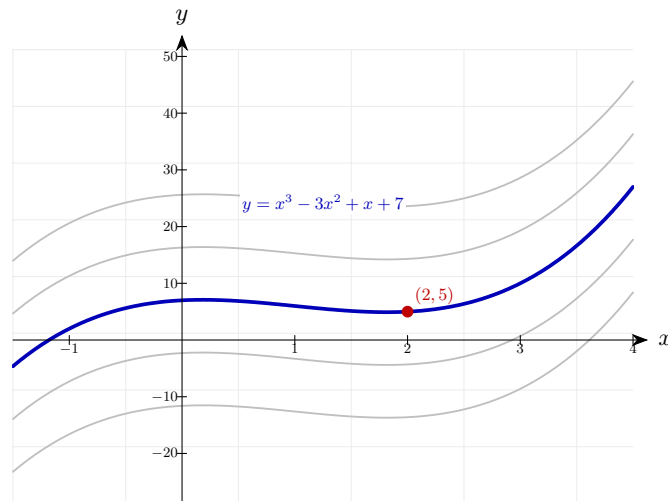
Step 3: State the equation.

$$f(x) = x^3 - 3x^2 + x + 7$$

Substitute $C = 7$.

Final answer: $f(x) = x^3 - 3x^2 + x + 7$

Diagram.



The family of antiderivatives (grey) with the particular solution (blue) through the given point.

Indefinite Integration 23 (i1-023)

Difficulty: Foundation / Marks: 4

A curve $y = f(x)$ passes through $(1, -2)$ and $f'(x) = 6x(x - 2)$. Find $f(x)$.

Worked solution.

Step 1: Expand $f'(x)$.

$$f'(x) = 6x^2 - 12x$$

Multiply $6x$ by each term in the bracket.

Step 2: Integrate.

$$f(x) = \int (6x^2 - 12x) dx = 2x^3 - 6x^2 + C$$

Apply the power rule.

Step 3: Use the point $(1, -2)$.

$$-2 = 2(1) - 6(1) + C = -4 + C \implies C = 2$$

Substitute $x = 1$, $f(1) = -2$.

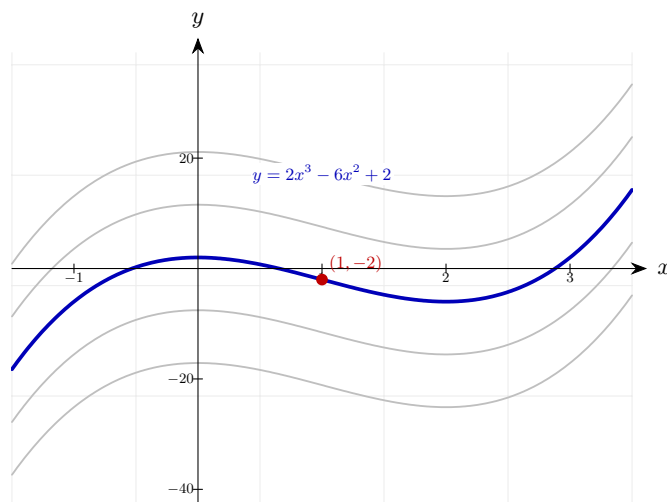
Step 4: State the answer.

$$f(x) = 2x^3 - 6x^2 + 2$$

Substitute $C = 2$.

Final answer: $f(x) = 2x^3 - 6x^2 + 2$

Diagram.



The family of antiderivatives (grey) with the particular solution (blue) through the given point.

Indefinite Integration 24 (i1-024)

Difficulty: Foundation / Marks: 4

The gradient of a curve is given by $\frac{dy}{dx} = \frac{1}{\sqrt{x}}$. Given that the curve passes through $(4, 10)$, find the equation of the curve.

Worked solution.

Step 1: Rewrite $\frac{1}{\sqrt{x}} = x^{-1/2}$.

$$y = \int x^{-1/2} dx$$

Convert to index notation.

Step 2: Integrate.

$$y = \frac{x^{1/2}}{1/2} + C = 2\sqrt{x} + C$$

Increase index from $-\frac{1}{2}$ to $\frac{1}{2}$ and divide by $\frac{1}{2}$.

Step 3: Use the point $(4, 10)$.

$$10 = 2\sqrt{4} + C = 4 + C \implies C = 6$$

Substitute $x = 4$, $y = 10$.

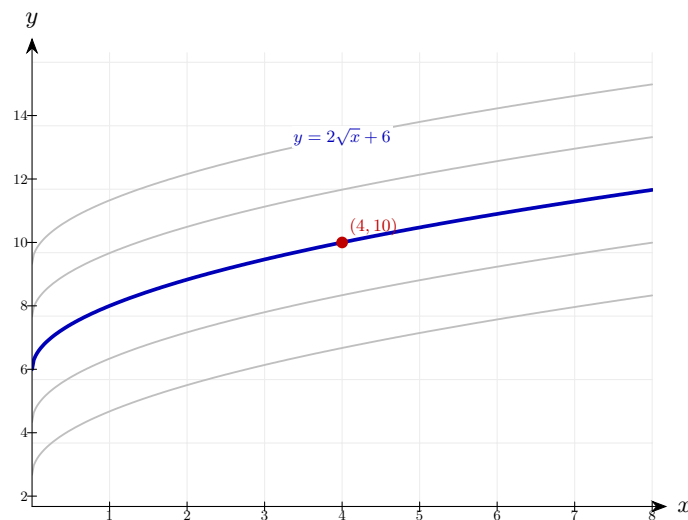
Step 4: Write the equation.

$$y = 2\sqrt{x} + 6$$

Substitute $C = 6$.

Final answer: $y = 2\sqrt{x} + 6$

Diagram.



The family of antiderivatives (grey) with the particular solution (blue) through the given point.

Indefinite Integration 25 (i1-025)

Difficulty: Foundation / Marks: 5

A curve passes through $(3, 4)$ and has gradient function $f'(x) = 2x - \frac{5}{x^2}$. Find $f(x)$.

Worked solution.

Step 1: Rewrite $\frac{5}{x^2} = 5x^{-2}$.

$$f'(x) = 2x - 5x^{-2}$$

Convert to index notation.

Step 2: Integrate.

$$f(x) = x^2 - 5 \cdot \frac{x^{-1}}{-1} + C = x^2 + \frac{5}{x} + C$$

Apply the power rule to each term.

Step 3: Use the point $(3, 4)$.

$$4 = 9 + \frac{5}{3} + C \implies C = 4 - 9 - \frac{5}{3} = -5 - \frac{5}{3} = -\frac{20}{3}$$

Substitute $x = 3$, $f(3) = 4$.

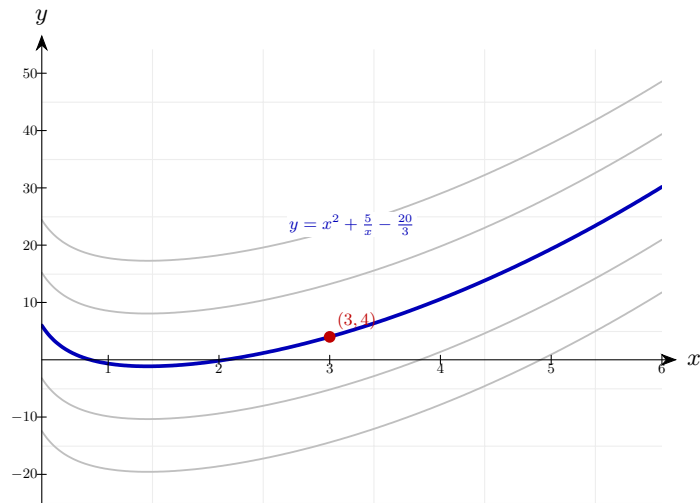
Step 4: State the equation.

$$f(x) = x^2 + \frac{5}{x} - \frac{20}{3}$$

Substitute $C = -\frac{20}{3}$.

Final answer: $f(x) = x^2 + \frac{5}{x} - \frac{20}{3}$

Diagram.



The family of antiderivatives (grey) with the particular solution (blue) through the given point.

Indefinite Integration 26 (i1-026)

Difficulty: Foundation / Marks: 4

Given that $\frac{dy}{dt} = \sqrt{t} - 3$ and $y = 9$ when $t = 4$, find y as a function of t .

Worked solution.

Step 1: Write $\sqrt{t} = t^{1/2}$ and integrate.

$$y = \int (t^{1/2} - 3) dt = \frac{2}{3}t^{3/2} - 3t + C$$

Apply the power rule to each term in t .

Step 2: Use $y = 9$ when $t = 4$.

$$9 = \frac{2}{3}(8) - 12 + C = \frac{16}{3} - 12 + C$$

$$4^{3/2} = (\sqrt{4})^3 = 8.$$

Step 3: Solve for C .

$$C = 9 - \frac{16}{3} + 12 = 21 - \frac{16}{3} = \frac{63 - 16}{3} = \frac{47}{3}$$

Rearrange and simplify.

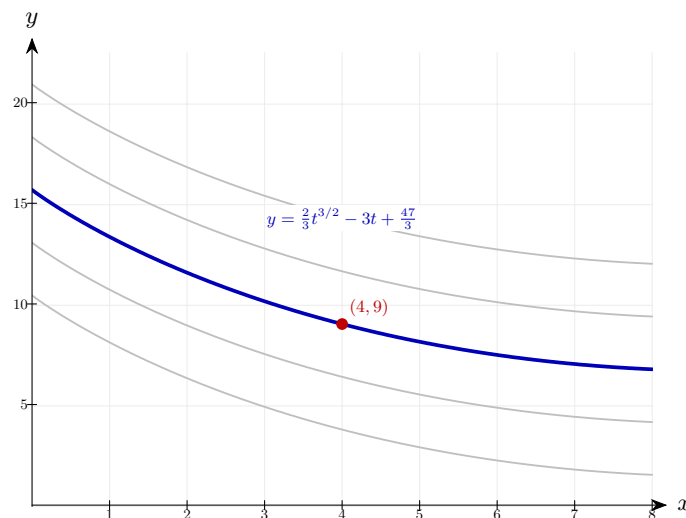
Step 4: State y .

$$y = \frac{2}{3}t^{3/2} - 3t + \frac{47}{3}$$

Substitute $C = \frac{47}{3}$.

$$\text{Final answer: } y = \frac{2}{3}t^{3/2} - 3t + \frac{47}{3}$$

Diagram.



The family of antiderivatives (grey) with the particular solution (blue) through the given point.

Indefinite Integration 27 (i1-027)

Difficulty: Foundation / Marks: 5

A curve $y = f(x)$ passes through $(1, 3)$ and $f'(x) = (2x + 1)(x - 3)$. Find $f(x)$.

Worked solution.

Step 1: Expand $f'(x)$.

$$(2x + 1)(x - 3) = 2x^2 - 6x + x - 3 = 2x^2 - 5x - 3$$

FOIL the brackets.

Step 2: Integrate.

$$f(x) = \frac{2x^3}{3} - \frac{5x^2}{2} - 3x + C$$

Power rule applied to each term.

Step 3: Use the point $(1, 3)$.

$$3 = \frac{2}{3} - \frac{5}{2} - 3 + C \implies C = 3 - \frac{2}{3} + \frac{5}{2} + 3 = 6 - \frac{4}{6} + \frac{15}{6} = 6 + \frac{11}{6} = \frac{47}{6}$$

Substitute $x = 1$, $f(1) = 3$ and solve for C .

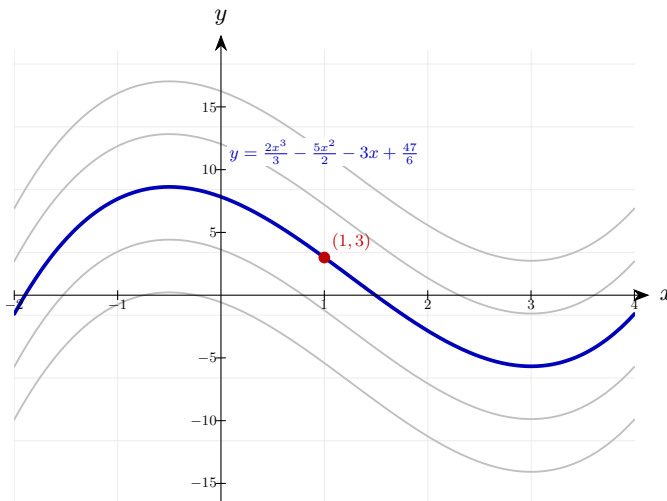
Step 4: Write the equation.

$$f(x) = \frac{2x^3}{3} - \frac{5x^2}{2} - 3x + \frac{47}{6}$$

Substitute $C = \frac{47}{6}$.

Final answer: $f(x) = \frac{2x^3}{3} - \frac{5x^2}{2} - 3x + \frac{47}{6}$

Diagram.



The family of antiderivatives (grey) with the particular solution (blue) through the given point.

Indefinite Integration 28 (i1-028)

Difficulty: Foundation / Marks: 5

A curve $y = f(x)$ passes through $(4, 9)$ and its gradient function is $f'(x) = \sqrt{x} + \frac{1}{\sqrt{x}}$. Find $f(x)$.

Worked solution.

Step 1: Rewrite in index notation.

$$f'(x) = x^{1/2} + x^{-1/2}$$

$$\sqrt{x} = x^{1/2} \text{ and } \frac{1}{\sqrt{x}} = x^{-1/2}.$$

Step 2: Integrate.

$$f(x) = \frac{x^{3/2}}{3/2} + \frac{x^{1/2}}{1/2} + C = \frac{2}{3}x^{3/2} + 2x^{1/2} + C$$

Apply the power rule to each term.

Step 3: Use the point $(4, 9)$.

$$9 = \frac{2}{3}(8) + 2(2) + C = \frac{16}{3} + 4 + C$$

$$4^{3/2} = 8 \text{ and } 4^{1/2} = 2.$$

Step 4: Solve for C .

$$C = 9 - \frac{16}{3} - 4 = 5 - \frac{16}{3} = \frac{15 - 16}{3} = -\frac{1}{3}$$

Rearrange.

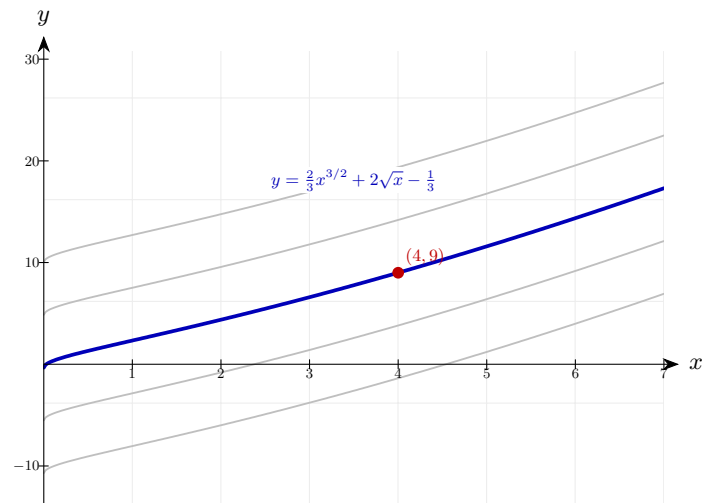
Step 5: State the equation.

$$f(x) = \frac{2}{3}x^{3/2} + 2\sqrt{x} - \frac{1}{3}$$

$$\text{Substitute } C = -\frac{1}{3}.$$

Final answer: $f(x) = \frac{2}{3}x^{3/2} + 2\sqrt{x} - \frac{1}{3}$

Diagram.



The family of antiderivatives (grey) with the particular solution (blue) through the given point.

Indefinite Integration 29 (i1-029)

Difficulty: Foundation / *Marks: 4*

An object moves so that its acceleration at time t seconds is $a = 6t - 2 \text{ ms}^{-2}$. Given that the object is at rest when $t = 0$, find the velocity v as a function of t .

Worked solution.

Step 1: Velocity is the integral of acceleration.

$$v = \int (6t - 2) dt = 3t^2 - 2t + C$$

Integrate with respect to t .

Step 2: Use $v = 0$ when $t = 0$.

$$0 = 0 - 0 + C \implies C = 0$$

Substitute the initial condition.

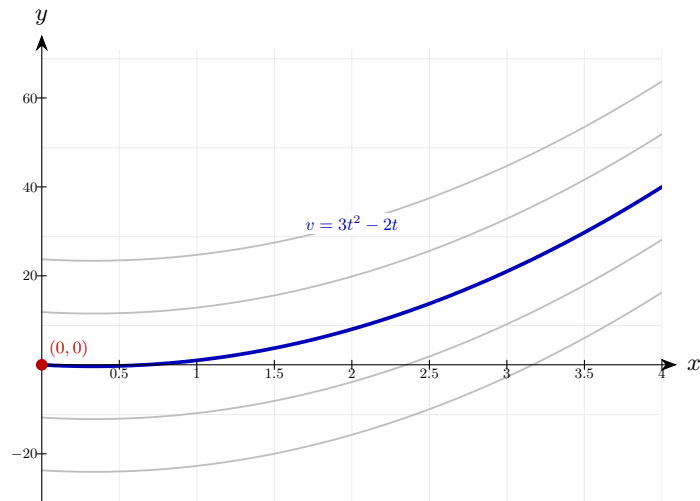
Step 3: State the velocity function.

$$v = 3t^2 - 2t$$

No additional constant is needed.

Final answer: $v = 3t^2 - 2t$

Diagram.



The family of antiderivatives (grey) with the particular solution (blue) through the given point.

Indefinite Integration 30 (i1-030)

Difficulty: Foundation / Marks: 5

An object has acceleration $a = 10 - 4t \text{ ms}^{-2}$. Given that the velocity is 8 ms^{-1} when $t = 0$, find: (a) The velocity function. (b) The time when the object reaches its maximum velocity.

Worked solution.

Step 1: (a) Integrate acceleration to find velocity.

$$v = \int (10 - 4t) dt = 10t - 2t^2 + C$$

Power rule.

Step 2: Apply initial condition $v(0) = 8$.

$$8 = 0 + C \implies C = 8$$

Substitute $t = 0$, $v = 8$.

Step 3: State velocity function.

$$v = 10t - 2t^2 + 8$$

Substitute $C = 8$.

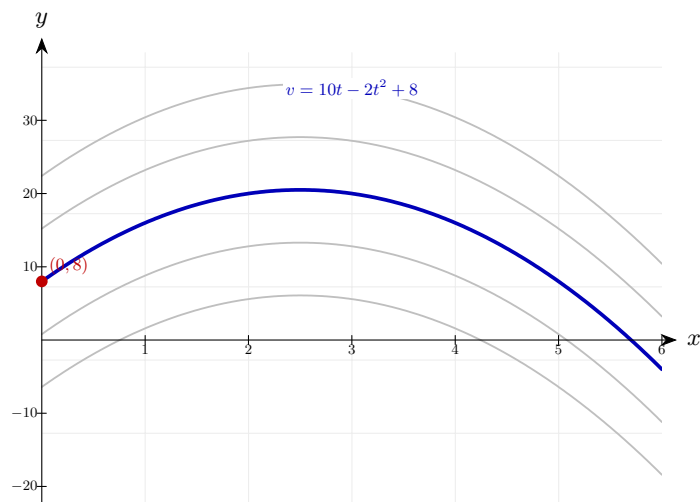
Step 4: (b) Maximum velocity when $a = \frac{dv}{dt} = 0$.

$$10 - 4t = 0 \implies t = 2.5 \text{ s}$$

Set the acceleration to zero.

Final answer: (a) $v = 10t - 2t^2 + 8$ \ (b) Maximum velocity at $t = 2.5 \text{ s}$

Diagram.



The family of antiderivatives (grey) with the particular solution (blue) through the given point.

Indefinite Integration 31 (i1-031)

Difficulty: Foundation / Marks: 4

Find $\int \left(\frac{x^3 + 2x^2 - x}{x^2} \right) dx$.

Worked solution.

Step 1: Divide each numerator term by x^2 .

$$x + 2 - x^{-1}$$

Splitting the fraction term-by-term gives a sum of powers of x . Using index laws: $\frac{x^3}{x^2} = x$, $\frac{2x^2}{x^2} = 2$, and $\frac{-x}{x^2} = -x^{-1}$.

Step 2: Set up the integral.

$$\int (x + 2 - x^{-1}) dx$$

The reverse power rule $\int x^n dx = \frac{x^{n+1}}{n+1} + C$ handles the x and the constant 2 directly, but it fails for x^{-1} because then $n+1 = 0$ and we would divide by zero — this is the famous $n = -1$ exclusion.

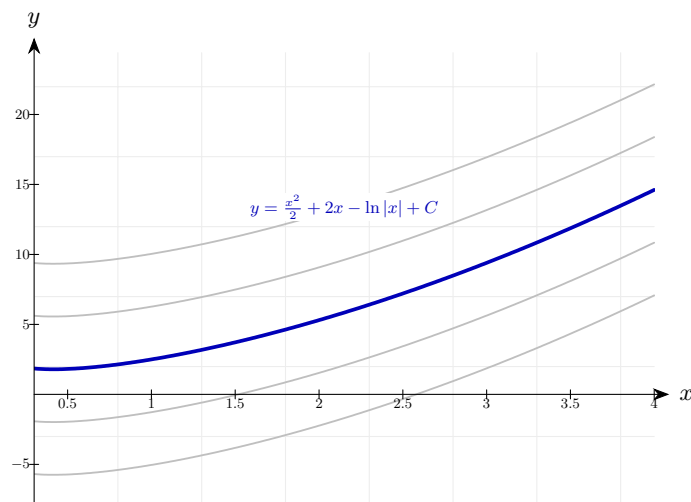
Step 3: Integrate, using the standard result $\int x^{-1} dx = \ln|x|$.

$$= \frac{x^2}{2} + 2x - \ln|x| + C$$

For the x^{-1} term we use the standard logarithm result $\int \frac{1}{x} dx = \ln|x| + C$ (the absolute value handles negative x). One $+C$ covers the whole expression — a single antiderivative family.

Final answer: $\frac{x^2}{2} + 2x - \ln|x| + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 32 (i1-032)

Difficulty: Foundation / Marks: 6

A function f has gradient $f'(x) = 3x^2 + kx - 4$, where k is a constant. Given that $f'(2) = 12$ and the curve passes through $(0, 5)$, find k and hence find $f(x)$.

Worked solution.

Step 1: Use $f'(2) = 12$ to find k .

$$3(2)^2 + k(2) - 4 = 12 \implies 12 + 2k - 4 = 12 \implies 2k = 4 \implies k = 2$$

Substitute $x = 2$ into $f'(x)$ and equate to the given gradient. This generates a linear equation in the unknown k , which we then solve.

Step 2: Integrate $f'(x) = 3x^2 + 2x - 4$.

$$f(x) = \int (3x^2 + 2x - 4) \, dx = x^3 + x^2 - 4x + C$$

Apply the reverse power rule $\int x^n \, dx = \frac{x^{n+1}}{n+1} + C$ to each term: $3 \cdot \frac{x^3}{3} = x^3$, $2 \cdot \frac{x^2}{2} = x^2$, and $\int -4 \, dx = -4x$. The $+C$ is essential because there is a whole family of antiderivatives that differ only by a constant.

Step 3: Use $f(0) = 5$ to pin down C .

$$5 = (0)^3 + (0)^2 - 4(0) + C \implies C = 5$$

The boundary condition selects one specific antiderivative from the family. Substituting $x = 0$ makes every x -term vanish, so the equation reduces directly to $C = 5$.

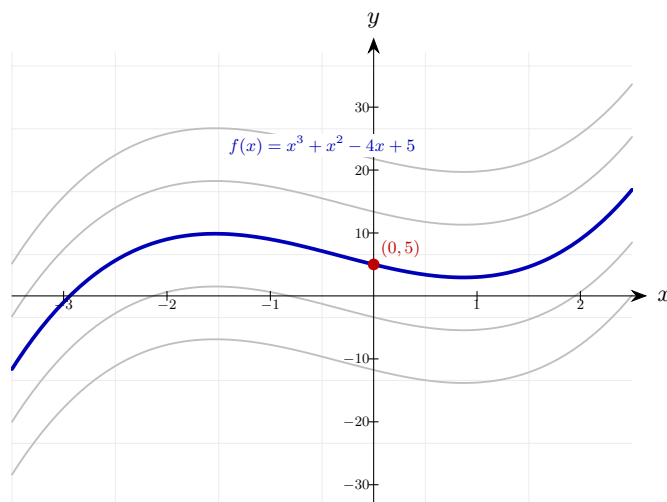
Step 4: State the answer.

$$f(x) = x^3 + x^2 - 4x + 5$$

Substitute the found value $C = 5$ back into the general antiderivative to get the unique $f(x)$ that satisfies both conditions.

Final answer: $k = 2; f(x) = x^3 + x^2 - 4x + 5$

Diagram.



The family of antiderivatives (grey) with the particular solution (blue) through the given point.

Indefinite Integration 33 (i1-033)

Difficulty: Foundation / Marks: 5

The curve $y = f(x)$ passes through $(1, -1)$ and has $f'(x) = (5x - 1)(x + 2)$. Find $f(x)$.

Worked solution.

Step 1: Expand $f'(x)$.

$$(5x - 1)(x + 2) = 5x^2 + 10x - x - 2 = 5x^2 + 9x - 2$$

FOIL the brackets.

Step 2: Integrate.

$$f(x) = \frac{5x^3}{3} + \frac{9x^2}{2} - 2x + C$$

Apply the power rule.

Step 3: Use $(1, -1)$.

$$-1 = \frac{5}{3} + \frac{9}{2} - 2 + C$$

Substitute $x = 1$.

Step 4: Solve for C .

$$C = -1 - \frac{5}{3} - \frac{9}{2} + 2 = 1 - \frac{10}{6} - \frac{27}{6} = 1 - \frac{37}{6} = -\frac{31}{6}$$

Common denominator of 6.

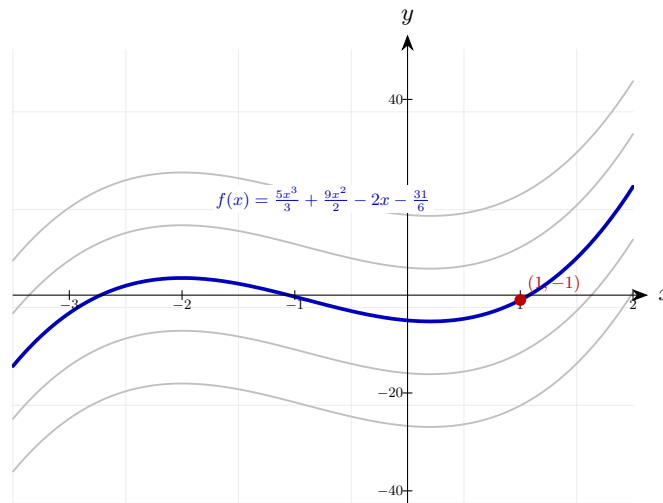
Step 5: State the equation.

$$f(x) = \frac{5x^3}{3} + \frac{9x^2}{2} - 2x - \frac{31}{6}$$

Substitute $C = -\frac{31}{6}$.

Final answer: $f(x) = \frac{5x^3}{3} + \frac{9x^2}{2} - 2x - \frac{31}{6}$

Diagram.



The family of antiderivatives (grey) with the particular solution (blue) through the given point.

Indefinite Integration 34 (i1-034)

Difficulty: Foundation / Marks: 5

A curve $y = f(x)$ has $f'(x) = x^2 - \frac{4}{x^2}$ and passes through $(2, 3)$. Find $f(x)$.

Worked solution.

Step 1: Rewrite $\frac{4}{x^2} = 4x^{-2}$.

$$f'(x) = x^2 - 4x^{-2}$$

Index notation.

Step 2: Integrate.

$$f(x) = \frac{x^3}{3} - 4 \cdot \frac{x^{-1}}{-1} + C = \frac{x^3}{3} + \frac{4}{x} + C$$

Apply the power rule to each term.

Step 3: Use $f(2) = 3$.

$$3 = \frac{8}{3} + \frac{4}{2} + C = \frac{8}{3} + 2 + C$$

Substitute $x = 2$.

Step 4: Solve for C .

$$C = 3 - \frac{8}{3} - 2 = 1 - \frac{8}{3} = -\frac{5}{3}$$

Rearrange.

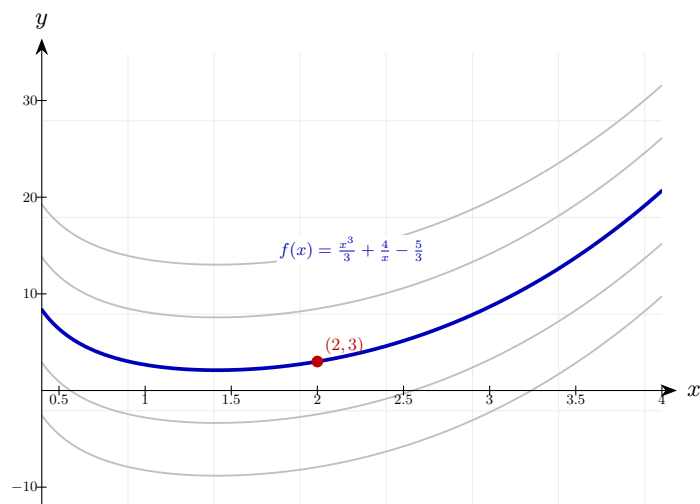
Step 5: State the equation.

$$f(x) = \frac{x^3}{3} + \frac{4}{x} - \frac{5}{3}$$

Substitute $C = -\frac{5}{3}$.

Final answer: $f(x) = \frac{x^3}{3} + \frac{4}{x} - \frac{5}{3}$

Diagram.



The family of antiderivatives (grey) with the particular solution (blue) through the given point.

Indefinite Integration 35 (i1-035)

Difficulty: Foundation / Marks: 7

A function f has gradient function $f'(x) = 4x^3 + 3kx^2 + k$, where k is a constant. The curve $y = f(x)$ passes through $(1, 6)$ with gradient 12 at that point. (a) Find the value of k . (b) Find $f(x)$.

Worked solution.

Step 1: (a) Use the gradient condition $f'(1) = 12$ to set up an equation in k .

$$f'(1) = 4(1)^3 + 3k(1)^2 + k = 4 + 3k + k = 4 + 4k$$

Substitute $x = 1$ into the gradient function and collect like terms. The unknown k survives because the two k -terms have the same sign, so they add rather than cancel.

Step 2: Set equal to 12 and solve.

$$4 + 4k = 12 \implies 4k = 8 \implies k = 2$$

A simple linear equation. Subtract 4 from both sides, then divide by 4.

Step 3: (b) Substitute $k = 2$ into $f'(x)$ before integrating.

$$f'(x) = 4x^3 + 6x^2 + 2$$

Replacing the unknown constant first keeps the antiderivative computation tidy.

Step 4: Integrate using the reverse power rule.

$$f(x) = \int (4x^3 + 6x^2 + 2) \, dx = x^4 + 2x^3 + 2x + C$$

Apply $\int x^n dx = \frac{x^{n+1}}{n+1} + C$ to each term: $4 \cdot \frac{x^4}{4} = x^4$, $6 \cdot \frac{x^3}{3} = 2x^3$, and $\int 2 dx = 2x$. The $+C$ is essential — without a boundary condition the antiderivative is only known up to an additive constant.

Step 5: Use the point (1,6) to pin down C.

$$6 = (1)^4 + 2(1)^3 + 2(1) + C = 1 + 2 + 2 + C = 5 + C \implies C = 1$$

The boundary condition selects the one antiderivative whose graph actually passes through (1,6) from the infinite family parametrised by C .

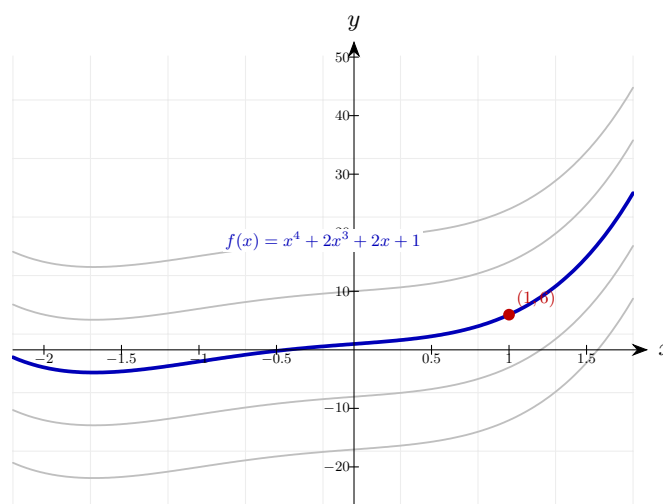
Step 6: State the equation.

$$f(x) = x^4 + 2x^3 + 2x + 1$$

Substitute $C = 1$ back into the general antiderivative.

Final answer: (a) $k = 2$ \ (b) $f(x) = x^4 + 2x^3 + 2x + 1$

Diagram.



The family of antiderivatives (grey) with the particular solution (blue) through the given point.

Indefinite Integration 36 (i1-036)

Difficulty: Foundation / Marks: 1

Find $\int 7x^2 dx$.

Worked solution.

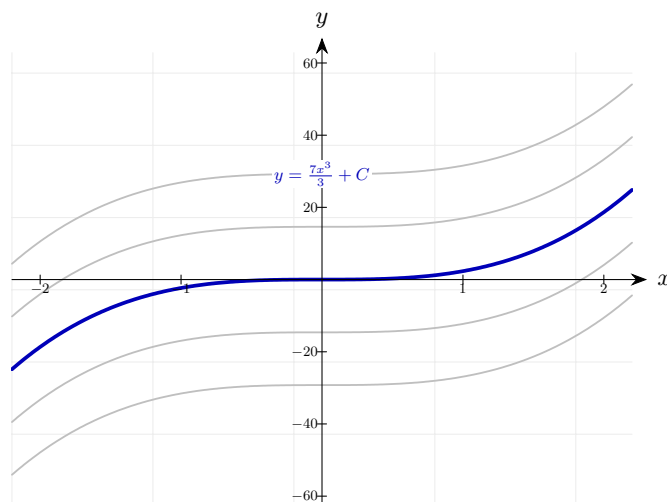
Apply the power rule.

$$\int 7x^2 dx = 7 \cdot \frac{x^3}{3} + C = \frac{7x^3}{3} + C$$

Increase the index from 2 to 3 and divide by 3.

Final answer: $\frac{7x^3}{3} + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 37 (i1-037)

Difficulty: Foundation / Marks: 2

Find $\int (3x^2 + 4x - 1) dx$.

Worked solution.

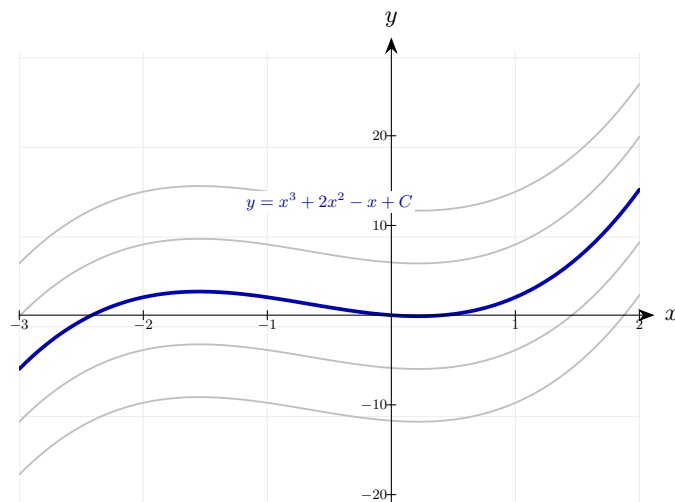
Integrate term by term.

$$\int (3x^2 + 4x - 1) dx = x^3 + 2x^2 - x + C$$

Apply the reverse power rule $\int x^n dx = \frac{x^{n+1}}{n+1} + C$ to each term: $3 \cdot \frac{x^3}{3} = x^3$, $4 \cdot \frac{x^2}{2} = 2x^2$, and integrating the constant -1 gives $-x$. One $+C$ covers the whole expression.

Final answer: $x^3 + 2x^2 - x + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 38 (i1-038)

Difficulty: Foundation / Marks: 1

Find $\int 10 \, dx$.

Worked solution.

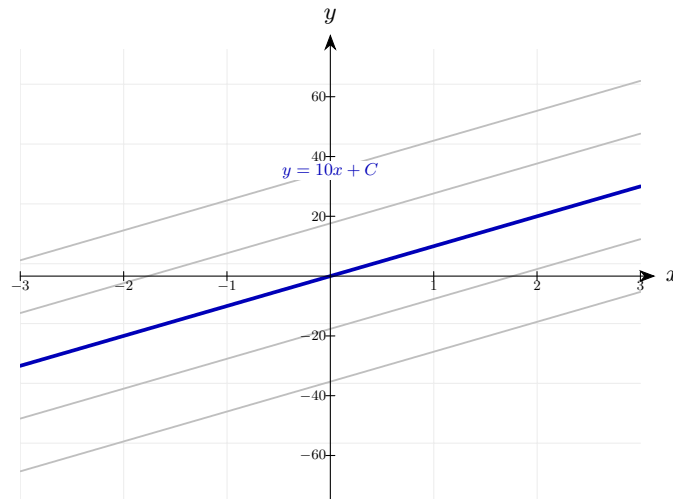
A constant integrates to the constant times x .

$$\int 10 \, dx = 10x + C$$

Think of 10 as $10x^0$; the power rule increases the index from 0 to 1 and divides by 1, giving $10x$. The $+C$ is required because every constant differentiates to zero, so all functions of the form $10x + C$ share the same derivative.

Final answer: $10x + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 39 (i1-039)

Difficulty: Foundation / Marks: 2

Find $\int x^{-3} \, dx$.

Worked solution.

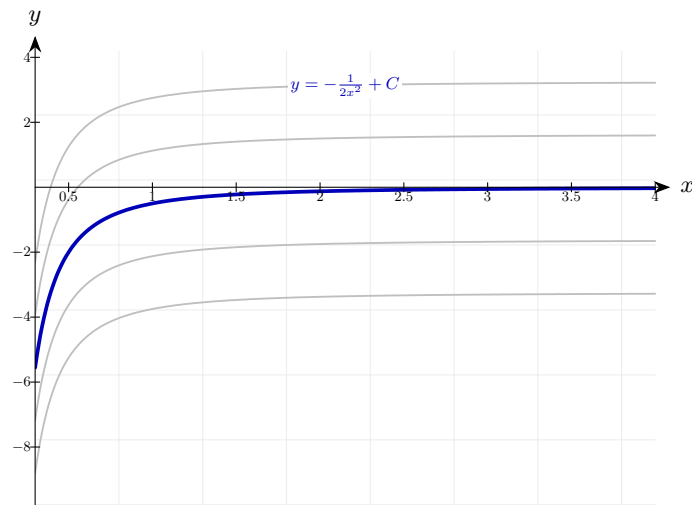
Apply the power rule with $n = -3$.

$$\int x^{-3} \, dx = \frac{x^{-2}}{-2} + C = -\frac{1}{2x^2} + C$$

Increase the index: $-3 + 1 = -2$.

Final answer: $-\frac{1}{2x^2} + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 40 (i1-040)

Difficulty: Foundation / Marks: 2

Find $\int x^{\frac{1}{2}} dx$.

Worked solution.

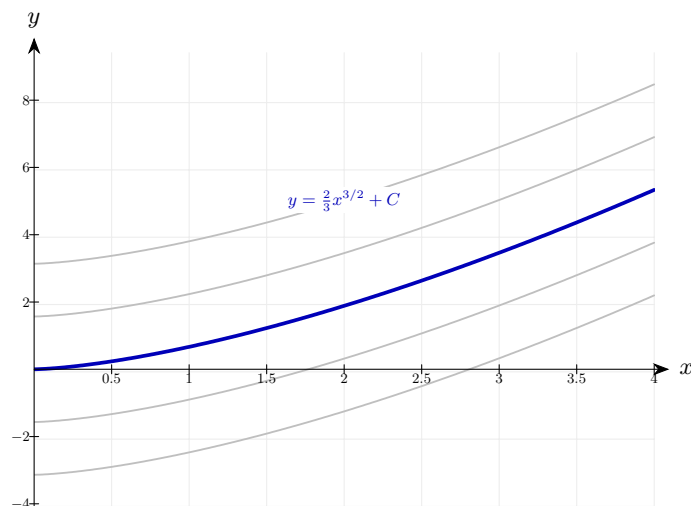
Apply the power rule with $n = \frac{1}{2}$.

$$\int x^{\frac{1}{2}} dx = \frac{x^{\frac{3}{2}}}{\frac{3}{2}} + C = \frac{2}{3}x^{\frac{3}{2}} + C$$

Increase the index: $\frac{1}{2} + 1 = \frac{3}{2}$. Dividing by $\frac{3}{2}$ is the same as multiplying by $\frac{2}{3}$.

Final answer: $\frac{2}{3}x^{\frac{3}{2}} + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 41 (i1-041)

Difficulty: Foundation / Marks: 2

Find $\int (6x^5 - 2x + 3) \, dx$.

Worked solution.

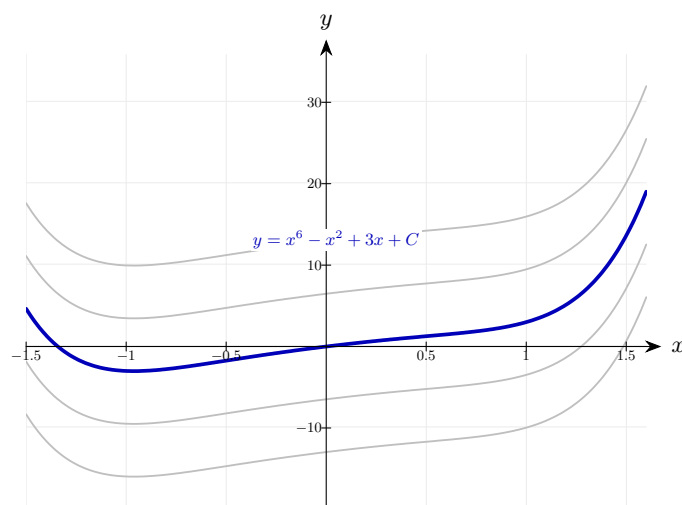
Integrate term by term.

$$\int (6x^5 - 2x + 3) \, dx = x^6 - x^2 + 3x + C$$

Apply the reverse power rule $\int x^n \, dx = \frac{x^{n+1}}{n+1} + C$ to each term: $6 \cdot \frac{x^6}{6} = x^6$, $2 \cdot \frac{x^2}{2} = x^2$, and $\int 3 \, dx = 3x$. One $+C$ covers the whole expression.

Final answer: $x^6 - x^2 + 3x + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 42 (i1-042)

Difficulty: Standard / Marks: 2

Find $\int \frac{3}{x^4} dx$.

Worked solution.

Step 1: Rewrite as a power of x .

$$\frac{3}{x^4} = 3x^{-4}$$

Convert the fraction to a negative-index form so the power rule applies cleanly. The rule $\int x^n dx = \frac{x^{n+1}}{n+1} + C$ works for any $n \neq -1$, including negative indices.

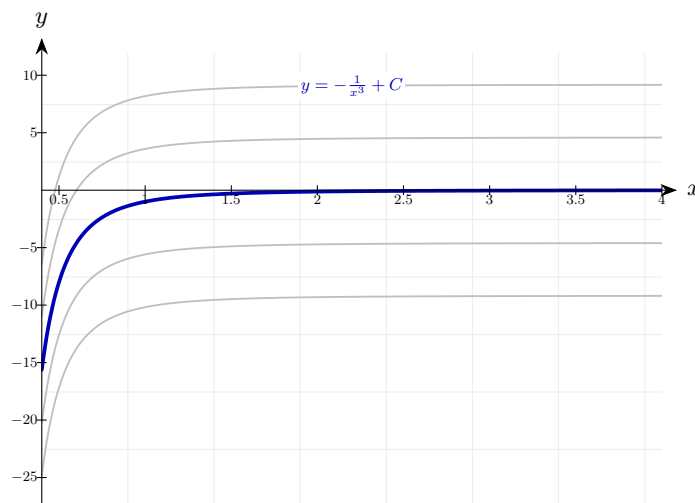
Step 2: Integrate.

$$\int 3x^{-4} dx = 3 \cdot \frac{x^{-3}}{-3} + C = -\frac{1}{x^3} + C$$

Increase the index from -4 to -3 and divide by -3 . The two 3s cancel, and dividing by a negative flips the sign — a common spot for sign errors. Always include $+C$.

Final answer: $-\frac{1}{x^3} + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 43 (i1-043)

Difficulty: Standard / Marks: 2

Find $\int 4\sqrt{x} dx$.

Worked solution.

Step 1: Rewrite the surd as a fractional power.

$$4\sqrt{x} = 4x^{\frac{1}{2}}$$

Always convert surds to fractional powers before integrating — the reverse power rule needs an explicit index n . Here $\sqrt{x} = x^{1/2}$.

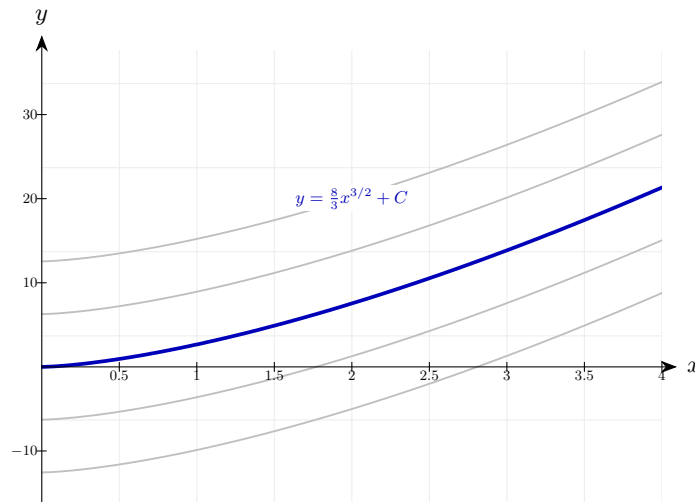
Step 2: Apply the power rule.

$$\int 4x^{\frac{1}{2}} dx = 4 \cdot \frac{x^{\frac{3}{2}}}{\frac{3}{2}} + C = \frac{8}{3}x^{\frac{3}{2}} + C$$

Increase the index from $\frac{1}{2}$ to $\frac{3}{2}$ and divide by $\frac{3}{2}$; dividing by a fraction is the same as multiplying by its reciprocal, so $4 \div \frac{3}{2} = 4 \times \frac{2}{3} = \frac{8}{3}$.

Final answer: $\frac{8}{3}x^{\frac{3}{2}} + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 44 (i1-044)

Difficulty: Standard / Marks: 3

Find $\int \left(x^3 - \frac{2}{x^2} \right) dx$.

Worked solution.

Step 1: Rewrite using a negative index.

$$x^3 - \frac{2}{x^2} = x^3 - 2x^{-2}$$

Converting $\frac{1}{x^2} = x^{-2}$ puts every term into the form x^n , so the power rule applies directly.

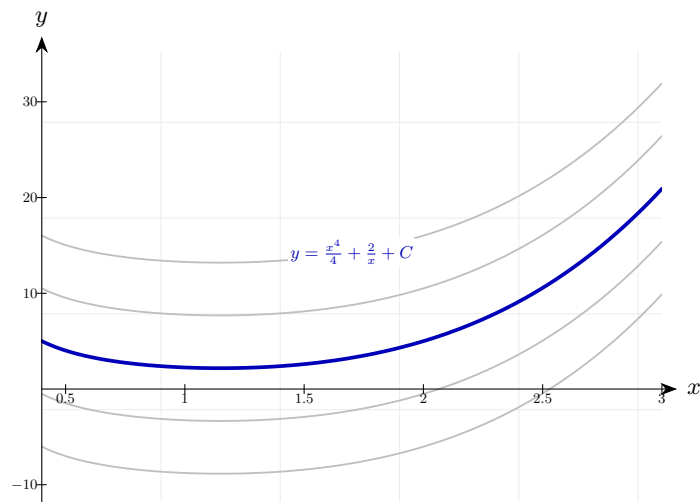
Step 2: Integrate.

$$\int (x^3 - 2x^{-2}) dx = \frac{x^4}{4} + \frac{2}{x} + C$$

For x^3 : increase the index to 4 and divide by 4. For $-2x^{-2}$: the new index is -1 , and dividing by -1 flips the sign giving $-2 \cdot \frac{x^{-1}}{-1} = \frac{2}{x}$ — watch the sign carefully.

Final answer: $\frac{x^4}{4} + \frac{2}{x} + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 45 (i1-045)

Difficulty: Standard / Marks: 3

Find $\int \frac{x^3 + 2x}{x} dx$.

Worked solution.

Step 1: Simplify the integrand by splitting the fraction.

$$\frac{x^3 + 2x}{x} = x^2 + 2$$

Divide each numerator term by x : $\frac{x^3}{x} = x^2$ and $\frac{2x}{x} = 2$. Simplifying before integrating avoids any need for the quotient rule or other heavy machinery — the power rule handles each term cleanly.

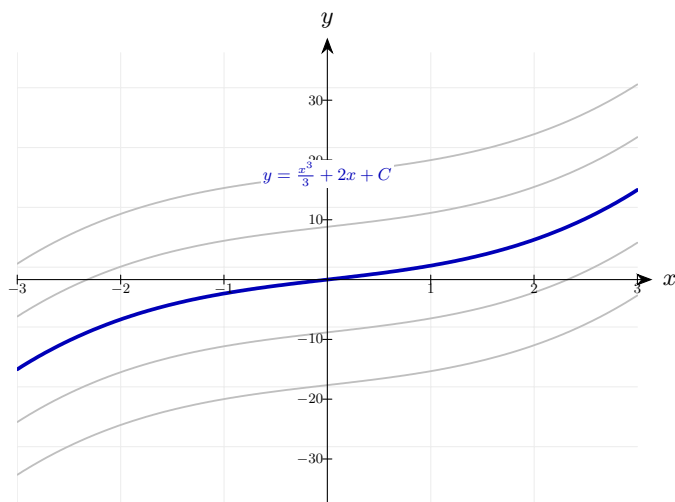
Step 2: Integrate term by term.

$$\int (x^2 + 2) dx = \frac{x^3}{3} + 2x + C$$

Apply the reverse power rule $\int x^n dx = \frac{x^{n+1}}{n+1} + C$: the x^2 term becomes $\frac{x^3}{3}$, and the constant 2 integrates to $2x$.

Final answer: $\frac{x^3}{3} + 2x + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 46 (i1-046)

Difficulty: Standard / Marks: 3

Find $\int \frac{6x^2 - 3}{x^2} dx$.

Worked solution.

Step 1: Split the fraction term-by-term.

$$\frac{6x^2 - 3}{x^2} = 6 - 3x^{-2}$$

$\frac{6x^2}{x^2} = 6$ and $\frac{-3}{x^2} = -3x^{-2}$. Writing the second term with a negative index puts it in the form x^n so the power rule can be applied.

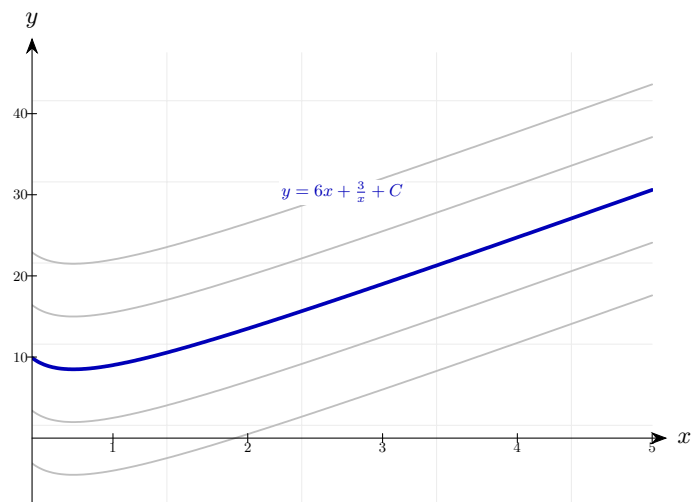
Step 2: Integrate.

$$\int (6 - 3x^{-2}) dx = 6x + \frac{3}{x} + C$$

Integrating the constant 6 gives $6x$. For $-3x^{-2}$: new index -1 , divide by -1 , giving $-3 \cdot \frac{x^{-1}}{-1} = +\frac{3}{x}$ — the two minus signs cancel, so be careful with the sign here.

Final answer: $6x + \frac{3}{x} + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 47 (i1-047)

Difficulty: Standard / Marks: 2

Find $\int x^{-\frac{1}{2}} dx$.

Worked solution.

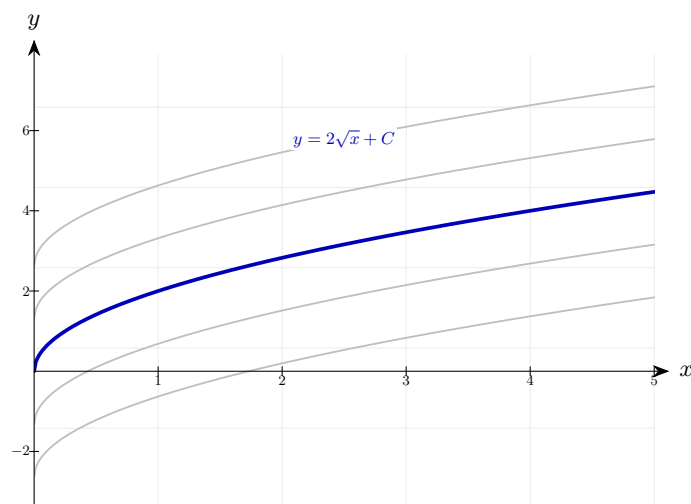
Apply the power rule.

$$\int x^{-\frac{1}{2}} dx = \frac{x^{\frac{1}{2}}}{\frac{1}{2}} + C = 2\sqrt{x} + C$$

$$-\frac{1}{2} + 1 = \frac{1}{2}.$$

Final answer: $2\sqrt{x} + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 48 (i1-048)

Difficulty: Standard / Marks: 3

Find $\int \left(2x + \frac{1}{\sqrt{x}}\right) dx$.

Worked solution.

Step 1: Rewrite the surd in index form.

$$2x + \frac{1}{\sqrt{x}} = 2x + x^{-\frac{1}{2}}$$

Converting $\frac{1}{\sqrt{x}} = x^{-1/2}$ puts the second term into a form the power rule can handle directly.

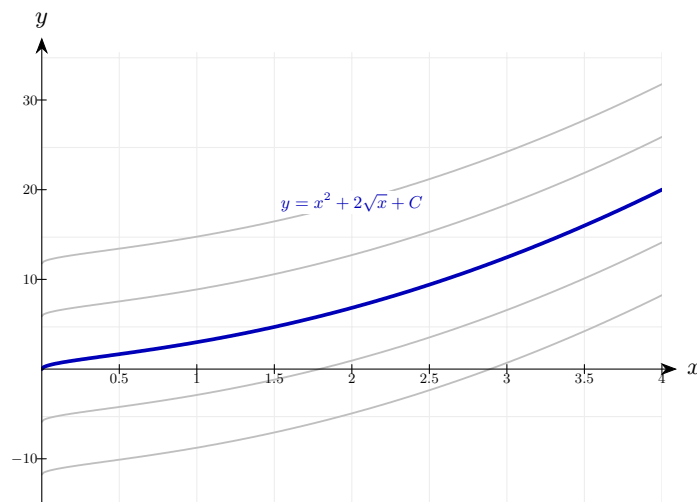
Step 2: Integrate term by term.

$$\int \left(2x + x^{-\frac{1}{2}}\right) dx = x^2 + 2x^{\frac{1}{2}} + C = x^2 + 2\sqrt{x} + C$$

For $2x$: the index becomes 2 and dividing by 2 cancels with the coefficient, giving x^2 . For $x^{-1/2}$: the new index is $\frac{1}{2}$, and dividing by $\frac{1}{2}$ doubles the term, giving $2\sqrt{x}$.

Final answer: $x^2 + 2\sqrt{x} + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 49 (i1-049)

Difficulty: Standard / Marks: 3

Find $\int (2x + 1)^2 dx$ by expanding first.

Worked solution.

Step 1: Expand the bracket.

$$(2x + 1)^2 = 4x^2 + 4x + 1$$

Use $(a + b)^2 = a^2 + 2ab + b^2$ with $a = 2x$, $b = 1$. You cannot integrate a bracket squared directly — expand first so each term is a separate power of x .

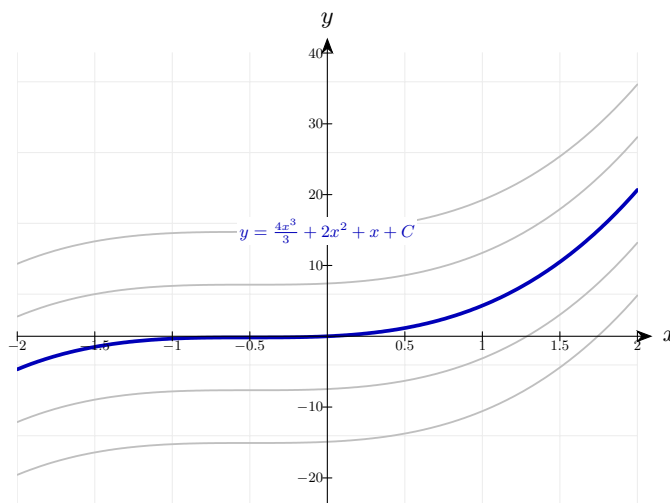
Step 2: Integrate term by term.

$$\int (4x^2 + 4x + 1) \, dx = \frac{4x^3}{3} + 2x^2 + x + C$$

Apply the reverse power rule to each term: $4 \cdot \frac{x^3}{3} = \frac{4x^3}{3}$, $4 \cdot \frac{x^2}{2} = 2x^2$, and $\int 1 \, dx = x$.

Final answer: $\frac{4x^3}{3} + 2x^2 + x + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 50 (i1-050)

Difficulty: Standard / Marks: 2

Find $\int x(x - 3) \, dx$.

Worked solution.

Step 1: Expand the bracket.

$$x(x - 3) = x^2 - 3x$$

Multiply x through the bracket. The product of two algebraic expressions cannot be integrated termwise unless it is first expanded into a sum.

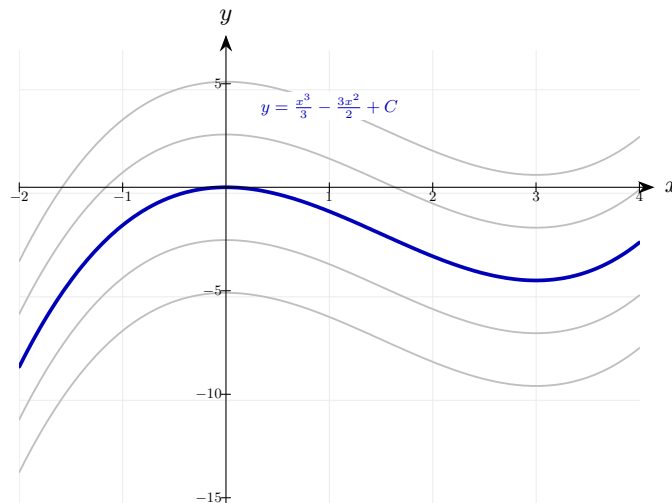
Step 2: Integrate.

$$\int (x^2 - 3x) \, dx = \frac{x^3}{3} - \frac{3x^2}{2} + C$$

Apply the reverse power rule to each term: $\int x^2 \, dx = \frac{x^3}{3}$ and $\int -3x \, dx = -3 \cdot \frac{x^2}{2} = -\frac{3x^2}{2}$.

Final answer: $\frac{x^3}{3} - \frac{3x^2}{2} + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 51 (i1-051)

Difficulty: Standard / Marks: 4

Find $\int \frac{x^2 + 4x + 3}{x + 1} dx$.

Worked solution.

Step 1: Factorise the numerator and cancel.

$$\frac{x^2 + 4x + 3}{x + 1} = \frac{(x + 1)(x + 3)}{x + 1} = x + 3$$

Find two numbers multiplying to 3 and adding to 4 — namely 1 and 3. Cancelling the common factor $(x + 1)$ reduces the quotient to a simple linear expression, which the power rule can handle directly.

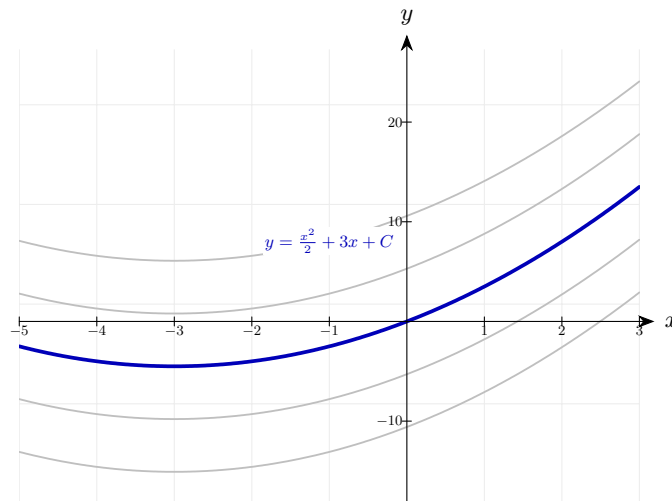
Step 2: Integrate.

$$\int (x + 3) dx = \frac{x^2}{2} + 3x + C$$

Apply the reverse power rule: $\int x dx = \frac{x^2}{2}$ and $\int 3 dx = 3x$.

Final answer: $\frac{x^2}{2} + 3x + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 52 (i1-052)

Difficulty: Standard / Marks: 3

Find $\int \sqrt{x}(x+4) \, dx$.

Worked solution.

Step 1: Expand by multiplying $x^{1/2}$ through the bracket.

$$\sqrt{x}(x+4) = x^{\frac{3}{2}} + 4x^{\frac{1}{2}}$$

Use index laws: $x^{1/2} \cdot x = x^{3/2}$ and $x^{1/2} \cdot 4 = 4x^{1/2}$. Converting $\sqrt{x}$ to $x^{1/2}$ first lets the power rule apply cleanly after expansion.

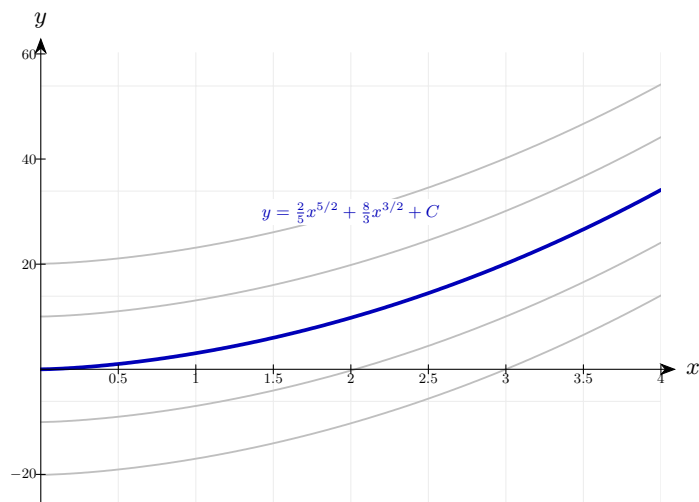
Step 2: Integrate each fractional-index term.

$$\int \left(x^{\frac{3}{2}} + 4x^{\frac{1}{2}} \right) dx = \frac{2}{5}x^{\frac{5}{2}} + \frac{8}{3}x^{\frac{3}{2}} + C$$

For $x^{3/2}$: new index $\frac{5}{2}$, divide by $\frac{5}{2}$ (i.e. multiply by $\frac{2}{5}$). For $4x^{1/2}$: new index $\frac{3}{2}$, then $4 \times \frac{2}{3} = \frac{8}{3}$.

Final answer: $\frac{2}{5}x^{\frac{5}{2}} + \frac{8}{3}x^{\frac{3}{2}} + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 53 (i1-053)

Difficulty: Standard / *Marks: 4*

Given that $\frac{dy}{dx} = 3x^2 - 4$ **and** $y = 5$ **when** $x = 2$, **find** y **in terms of** x .

Worked solution.

Step 1: Integrate the gradient function.

$$y = x^3 - 4x + C$$

Apply the reverse power rule: $3 \cdot \frac{x^3}{3} = x^3$ and $\int -4 \, dx = -4x$. The $+C$ is essential because integrating $\frac{dy}{dx}$ recovers y only up to an additive constant.

Step 2: Use the condition $y = 5$ when $x = 2$ to find C .

$$5 = (2)^3 - 4(2) + C = 8 - 8 + C \implies C = 5$$

The boundary condition pins down the value of C , selecting one specific curve from the family of antiderivatives that all share the same gradient function.

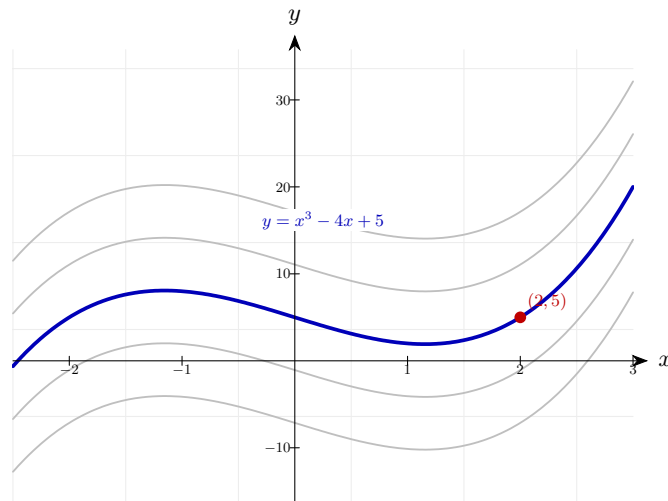
Step 3: State the equation of the curve.

$$y = x^3 - 4x + 5$$

Substitute $C = 5$ back into the general antiderivative.

Final answer: $y = x^3 - 4x + 5$

Diagram.



The family of antiderivatives (grey) with the particular solution (blue) through the given point.

Indefinite Integration 54 (i1-054)

Difficulty: Standard / Marks: 4

The gradient of a curve is $\frac{dy}{dx} = 6x - 1$. The curve passes through $(1, 4)$. Find the equation of the curve.

Worked solution.

Step 1: Integrate the gradient function.

$$y = 3x^2 - x + C$$

Apply the reverse power rule: $6 \cdot \frac{x^2}{2} = 3x^2$ and $\int -1 dx = -x$. The $+C$ accounts for the infinite family of curves with the same gradient.

Step 2: Substitute the point $(1, 4)$ to find C .

$$4 = 3(1)^2 - 1 + C = 3 - 1 + C \implies C = 2$$

The boundary condition picks out the one curve in the family that actually passes through $(1, 4)$.

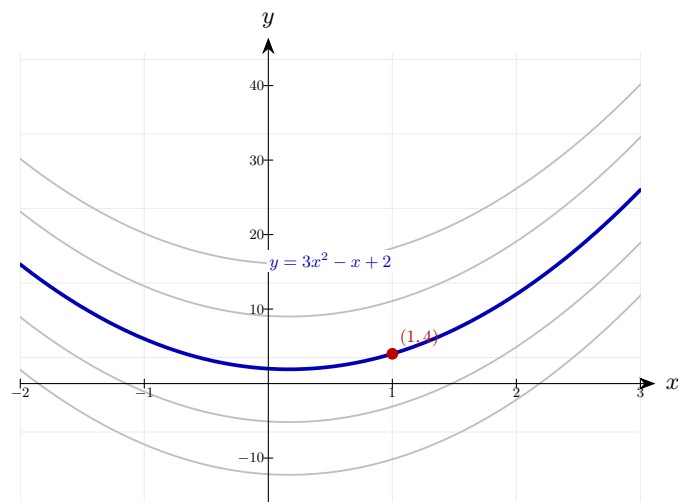
Step 3: State the equation.

$$y = 3x^2 - x + 2$$

Substitute $C = 2$ back into the general antiderivative.

Final answer: $y = 3x^2 - x + 2$

Diagram.



The family of antiderivatives (grey) with the particular solution (blue) through the given point.

Indefinite Integration 55 (i1-055)

Difficulty: Standard / Marks: 3

Find $\int \left(5x^4 + \frac{3}{x^2} - 7 \right) dx$.

Worked solution.

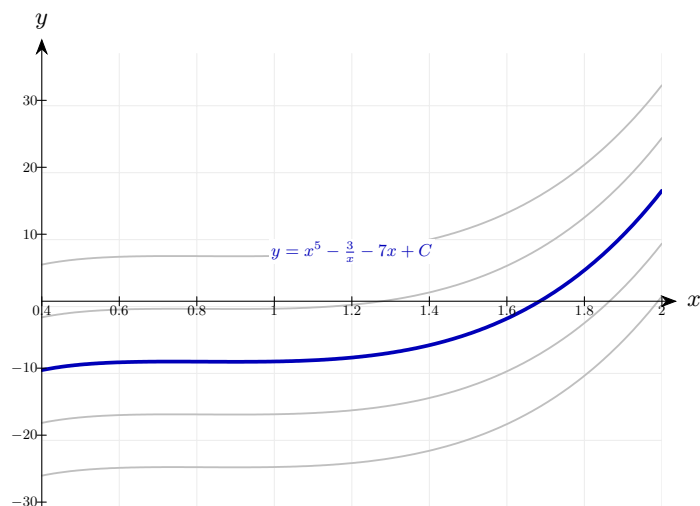
Rewrite $\frac{3}{x^2}$ as $3x^{-2}$ and integrate term by term.

$$\int (5x^4 + 3x^{-2} - 7) dx = x^5 - \frac{3}{x} - 7x + C$$

Converting to a negative index unlocks the power rule for the middle term. Integrating: $5 \cdot \frac{x^5}{5} = x^5$; for $3x^{-2}$, new index -1 and dividing by -1 flips the sign to give $-\frac{3}{x}$; and $\int -7 dx = -7x$. Watch the sign on the $-\frac{3}{x}$ carefully.

Final answer: $x^5 - \frac{3}{x} - 7x + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 56 (i1-056)

Difficulty: Standard / Marks: 4

Find $\int \frac{2x^4 - x^2 + 4}{x^2} dx$.

Worked solution.

Step 1: Divide each term in the numerator by x^2 .

$$\frac{2x^4 - x^2 + 4}{x^2} = 2x^2 - 1 + 4x^{-2}$$

Splitting the fraction term-by-term turns the awkward quotient into a sum of simple power-of- x terms, which the reverse power rule can handle directly. Note $\frac{4}{x^2} = 4x^{-2}$ — converting to a negative index is the key step that unlocks the power rule.

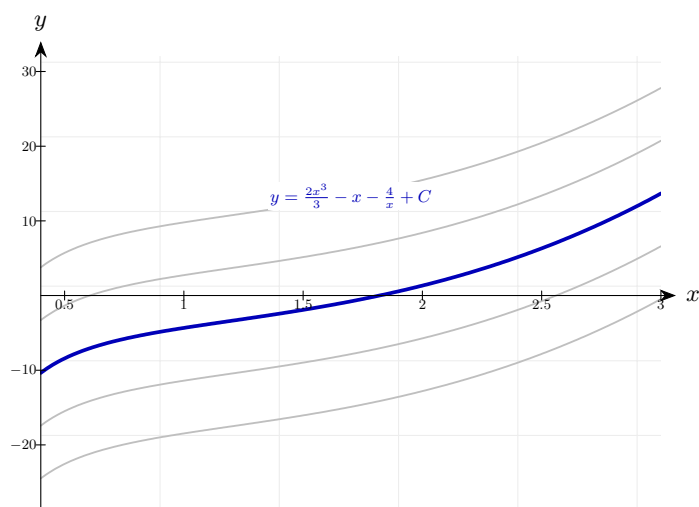
Step 2: Integrate using the reverse power rule.

$$\int (2x^2 - 1 + 4x^{-2}) dx = \frac{2x^3}{3} - x - \frac{4}{x} + C$$

Apply $\int x^n dx = \frac{x^{n+1}}{n+1} + C$ term-by-term. For $4x^{-2}$: the new index is -1 , and dividing by -1 flips the sign, giving $-4x^{-1} = -\frac{4}{x}$. The $+C$ is essential because indefinite integrals describe a whole family of antiderivatives.

Final answer: $\frac{2x^3}{3} - x - \frac{4}{x} + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 57 (i1-057)

Difficulty: Standard / Marks: 2

Find $\int x^{\frac{2}{3}} dx$.

Worked solution.

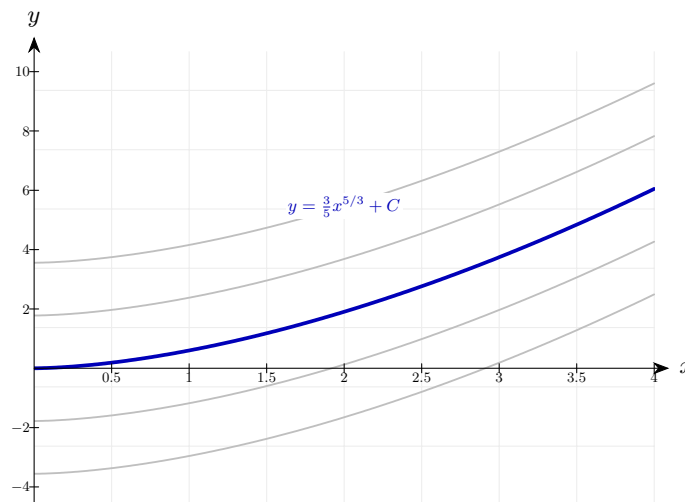
Apply the power rule.

$$\int x^{\frac{2}{3}} dx = \frac{x^{\frac{5}{3}}}{\frac{5}{3}} + C = \frac{3}{5}x^{\frac{5}{3}} + C$$

Increase the index by 1: $\frac{2}{3} + 1 = \frac{5}{3}$. Dividing by $\frac{5}{3}$ is the same as multiplying by its reciprocal $\frac{3}{5}$.

Final answer: $\frac{3}{5}x^{\frac{5}{3}} + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 58 (i1-058)

Difficulty: Standard / Marks: 2

Find $\int \frac{5}{\sqrt{x}} dx$.

Worked solution.

Step 1: Rewrite the surd in index form.

$$\frac{5}{\sqrt{x}} = 5x^{-\frac{1}{2}}$$

Using $\sqrt{x} = x^{1/2}$ and the reciprocal rule $\frac{1}{x^{1/2}} = x^{-1/2}$. Putting the expression in the form kx^n is essential before applying the power rule.

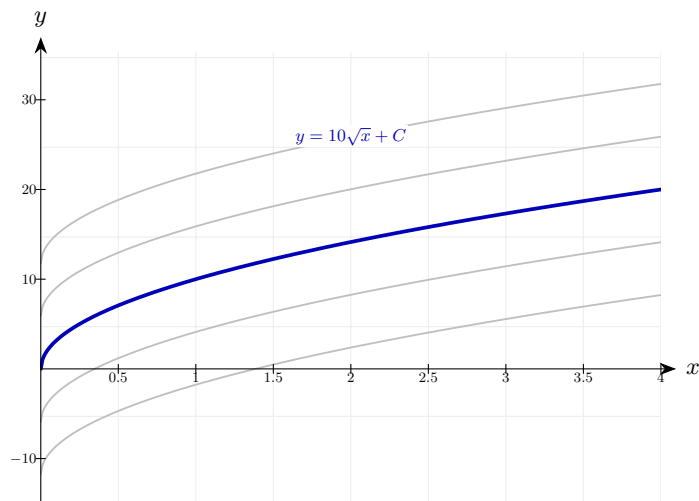
Step 2: Integrate.

$$\int 5x^{-\frac{1}{2}} dx = 5 \cdot 2x^{\frac{1}{2}} + C = 10\sqrt{x} + C$$

Increase the index from $-\frac{1}{2}$ to $\frac{1}{2}$ and divide by $\frac{1}{2}$, which is the same as multiplying by 2. Then $5 \times 2 = 10$, and $x^{1/2} = \sqrt{x}$.

Final answer: $10\sqrt{x} + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 59 (i1-059)

Difficulty: Standard / Marks: 3

Find $\int (x+2)(x-5) \, dx$.

Worked solution.

Step 1: Expand the brackets.

$$(x+2)(x-5) = x^2 - 3x - 10$$

FOIL: $x \cdot x - 5x + 2x - 10 = x^2 - 3x - 10$. Expand before integrating — the power rule applies to each individual power of x , not to a product of brackets.

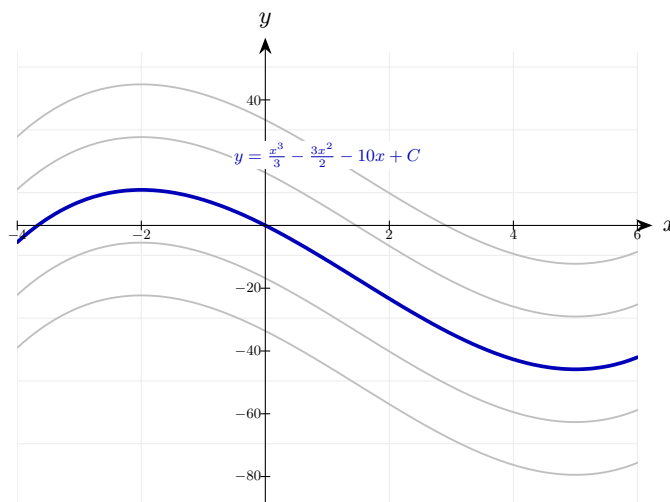
Step 2: Integrate term by term.

$$\int (x^2 - 3x - 10) \, dx = \frac{x^3}{3} - \frac{3x^2}{2} - 10x + C$$

Apply the reverse power rule: $\int x^2 \, dx = \frac{x^3}{3}$, $\int -3x \, dx = -\frac{3x^2}{2}$, and $\int -10 \, dx = -10x$.

Final answer: $\frac{x^3}{3} - \frac{3x^2}{2} - 10x + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 60 (i1-060)

Difficulty: Standard / *Marks: 4*

The gradient of a curve is $\frac{dy}{dx} = 4x^3 + 2x$. The curve passes through $(1, 0)$. Find the equation of the curve.

Worked solution.

Step 1: Integrate the gradient function.

$$y = x^4 + x^2 + C$$

Apply the reverse power rule: $4 \cdot \frac{x^4}{4} = x^4$ and $2 \cdot \frac{x^2}{2} = x^2$. The $+C$ is required because the gradient alone defines a whole family of parallel curves.

Step 2: Substitute $(1, 0)$ to find C .

$$0 = (1)^4 + (1)^2 + C = 1 + 1 + C \implies C = -2$$

The boundary condition selects the unique curve from the family that passes through this point.

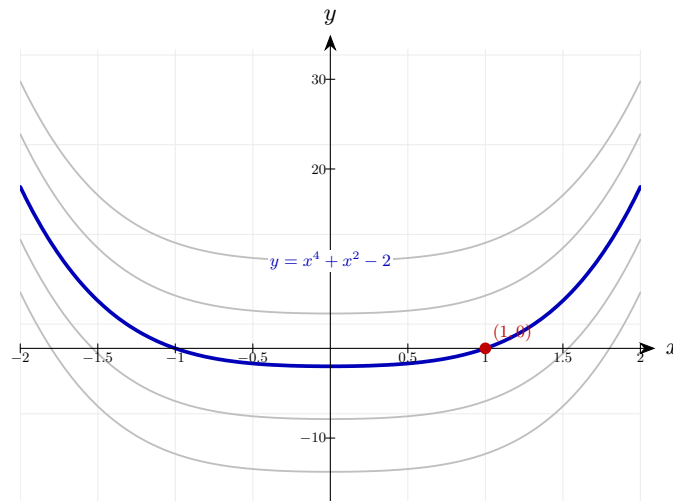
Step 3: State the equation.

$$y = x^4 + x^2 - 2$$

Substitute $C = -2$ into the general antiderivative.

Final answer: $y = x^4 + x^2 - 2$

Diagram.



The family of antiderivatives (grey) with the particular solution (blue) through the given point.

Indefinite Integration 61 (i1-061)

Difficulty: Standard / Marks: 3

Find $\int \left(3\sqrt{x} - \frac{1}{x^3} \right) dx$.

Worked solution.

Step 1: Rewrite in index form.

$$3\sqrt{x} - \frac{1}{x^3} = 3x^{\frac{1}{2}} - x^{-3}$$

Use $\sqrt{x} = x^{1/2}$ and $\frac{1}{x^3} = x^{-3}$. Putting every term as a power of x is essential before applying the power rule.

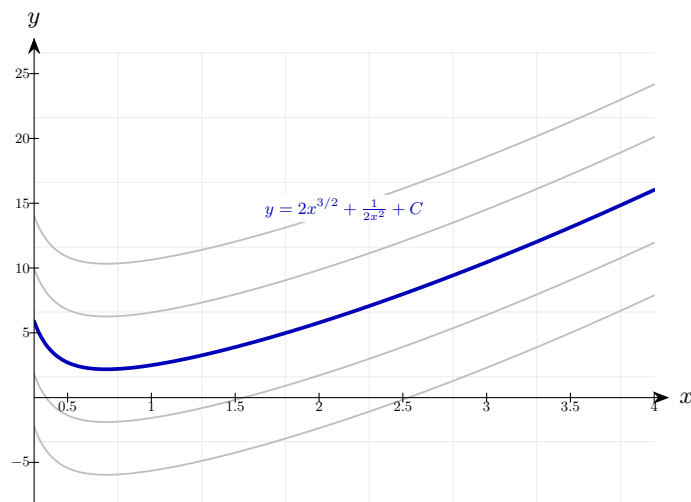
Step 2: Integrate each term.

$$\int \left(3x^{\frac{1}{2}} - x^{-3} \right) dx = 2x^{\frac{3}{2}} + \frac{1}{2x^2} + C$$

For $3x^{1/2}$: new index $\frac{3}{2}$, and $3 \times \frac{2}{3} = 2$. For $-x^{-3}$: new index -2 , divide by -2 , giving $-\frac{x^{-2}}{-2} = \frac{1}{2x^2}$ — the two minus signs cancel.

Final answer: $2x^{\frac{3}{2}} + \frac{1}{2x^2} + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 62 (i1-062)

Difficulty: Standard / *Marks: 4*

Given $f'(x) = 12x^2 - 6x + 1$ **and** $f(0) = 3$, **find** $f(x)$.

Worked solution.

Step 1: Integrate the gradient function.

$$f(x) = 4x^3 - 3x^2 + x + C$$

Apply the reverse power rule: $12 \cdot \frac{x^3}{3} = 4x^3$, $6 \cdot \frac{x^2}{2} = 3x^2$, and $\int 1 \, dx = x$. The $+C$ accounts for the unknown vertical shift.

Step 2: Use $f(0) = 3$ to pin down C .

$$3 = 4(0)^3 - 3(0)^2 + 0 + C \implies C = 3$$

Substituting $x = 0$ makes every x -term vanish, so the boundary value goes straight to C .

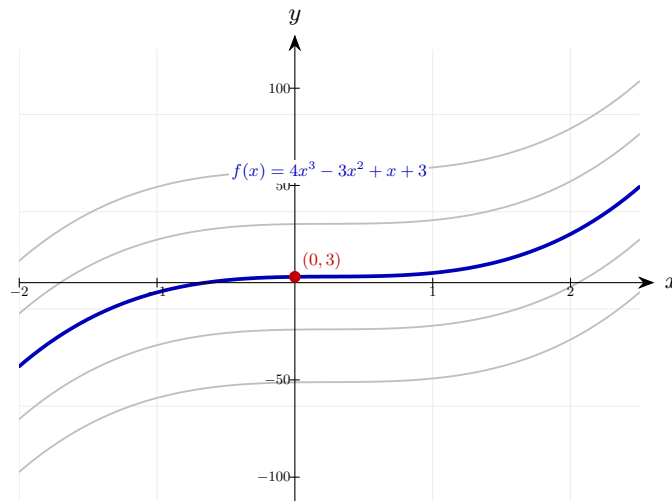
Step 3: State the final function.

$$f(x) = 4x^3 - 3x^2 + x + 3$$

Substitute $C = 3$ into the general antiderivative.

Final answer: $f(x) = 4x^3 - 3x^2 + x + 3$

Diagram.



The family of antiderivatives (grey) with the particular solution (blue) through the given point.

Indefinite Integration 63 (i1-063)

Difficulty: Standard / Marks: 4

Find $\int \frac{(x+1)(x-1)}{x^2} dx$.

Worked solution.

Step 1: Simplify using the difference of squares.

$$\frac{(x+1)(x-1)}{x^2} = \frac{x^2 - 1}{x^2} = 1 - x^{-2}$$

$(x+1)(x-1) = x^2 - 1$ by the difference of two squares. Then split the fraction: $\frac{x^2}{x^2} = 1$ and $\frac{-1}{x^2} = -x^{-2}$. Putting it in this form lets the power rule apply to each term.

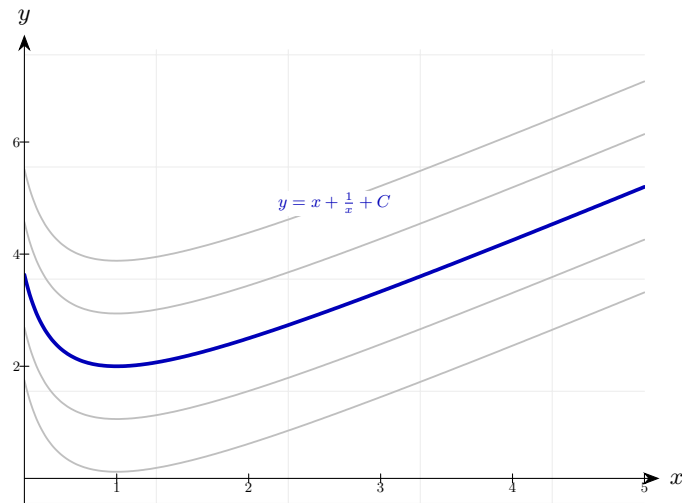
Step 2: Integrate.

$$\int (1 - x^{-2}) dx = x + \frac{1}{x} + C$$

$\int 1 dx = x$. For $-x^{-2}$: new index -1 , divide by -1 , giving $-\frac{x^{-1}}{-1} = +\frac{1}{x}$ — the two negatives cancel, so the answer has a plus sign.

Final answer: $x + \frac{1}{x} + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 64 (i1-064)

Difficulty: Standard / Marks: 4

Find $\int (x^2 + 1)^2 \, dx$.

Worked solution.

Step 1: Expand the bracket.

$$(x^2 + 1)^2 = x^4 + 2x^2 + 1$$

Use $(a + b)^2 = a^2 + 2ab + b^2$ with $a = x^2$, $b = 1$. The power rule cannot be applied to a bracket squared directly — expand into a sum first.

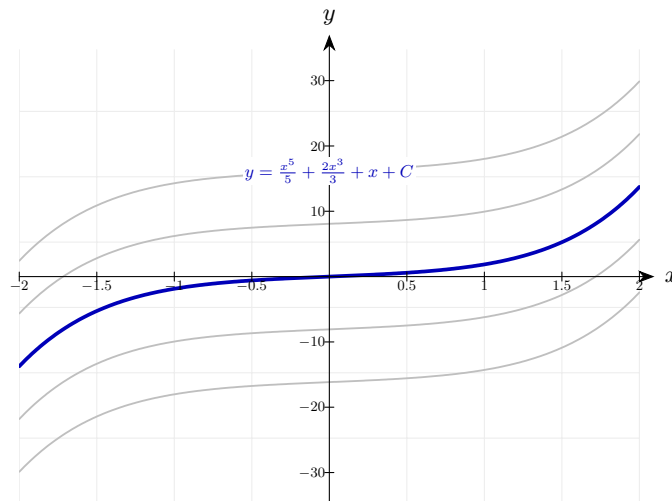
Step 2: Integrate term by term.

$$\int (x^4 + 2x^2 + 1) \, dx = \frac{x^5}{5} + \frac{2x^3}{3} + x + C$$

Apply the reverse power rule to each term: $\int x^4 \, dx = \frac{x^5}{5}$, $\int 2x^2 \, dx = \frac{2x^3}{3}$, and $\int 1 \, dx = x$.

Final answer: $\frac{x^5}{5} + \frac{2x^3}{3} + x + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 65 (i1-065)

Difficulty: Standard / Marks: 5

A curve has gradient function $\frac{dy}{dx} = \frac{3}{x^2} + 2$ and passes through $(3, 10)$. Find y .

Worked solution.

Step 1: Rewrite $\frac{3}{x^2}$ as $3x^{-2}$ and integrate.

$$y = \int (3x^{-2} + 2) dx = -\frac{3}{x} + 2x + C$$

For $3x^{-2}$: new index -1 , divide by -1 — this flips the sign to give $-\frac{3}{x}$. Then $\int 2 dx = 2x$. The $+C$ is required since the gradient alone does not fix the curve uniquely.

Step 2: Substitute the point $(3, 10)$ to find C .

$$10 = -\frac{3}{3} + 2(3) + C = -1 + 6 + C \implies C = 5$$

The boundary condition selects the unique curve from the family that passes through $(3, 10)$.

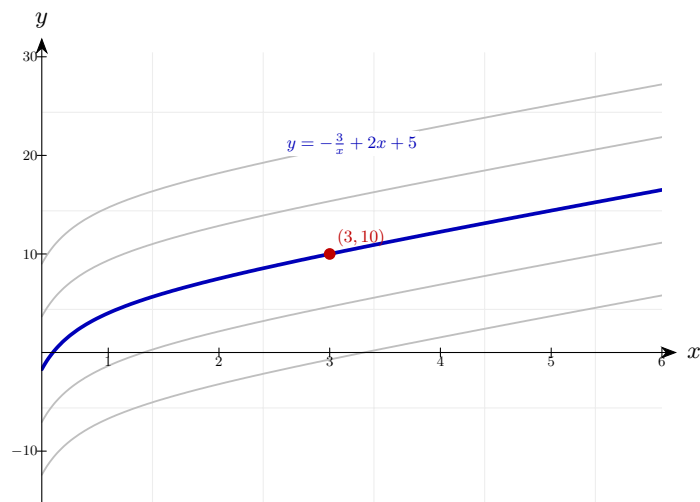
Step 3: State the equation.

$$y = -\frac{3}{x} + 2x + 5$$

Substitute $C = 5$ into the general antiderivative.

Final answer: $y = -\frac{3}{x} + 2x + 5$

Diagram.



The family of antiderivatives (grey) with the particular solution (blue) through the given point.

Indefinite Integration 66 (i1-066)

Difficulty: Standard / Marks: 3

Find $\int \frac{x^4 - 1}{x^2} dx$.

Worked solution.

Step 1: Split the fraction term-by-term.

$$\frac{x^4 - 1}{x^2} = x^2 - x^{-2}$$

$\frac{x^4}{x^2} = x^2$ and $\frac{-1}{x^2} = -x^{-2}$. Writing the second term with a negative index unlocks the power rule.

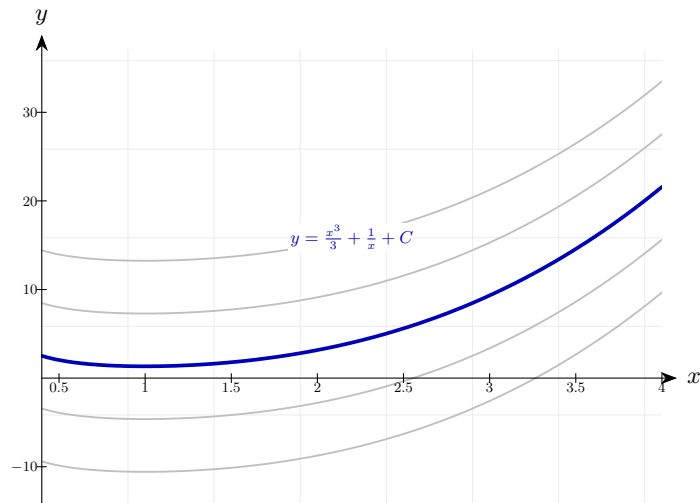
Step 2: Integrate.

$$\int (x^2 - x^{-2}) dx = \frac{x^3}{3} + \frac{1}{x} + C$$

$\int x^2 dx = \frac{x^3}{3}$. For $-x^{-2}$: new index -1 , divide by -1 , giving $-\frac{x^{-1}}{-1} = +\frac{1}{x}$ — the two minus signs cancel.

Final answer: $\frac{x^3}{3} + \frac{1}{x} + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 67 (i1-067)

Difficulty: Standard / Marks: 5

Find $\int \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right)^2 dx$.

Worked solution.

Step 1: Expand the bracket using $(a + b)^2 = a^2 + 2ab + b^2$.

$$\left(\sqrt{x} + \frac{1}{\sqrt{x}} \right)^2 = x + 2 + \frac{1}{x}$$

Here $a^2 = (\sqrt{x})^2 = x$, $b^2 = \frac{1}{x}$, and the middle term simplifies because $2 \cdot \sqrt{x} \cdot \frac{1}{\sqrt{x}} = 2$. Expanding before integrating is essential — the power rule cannot be applied directly to a bracket squared.

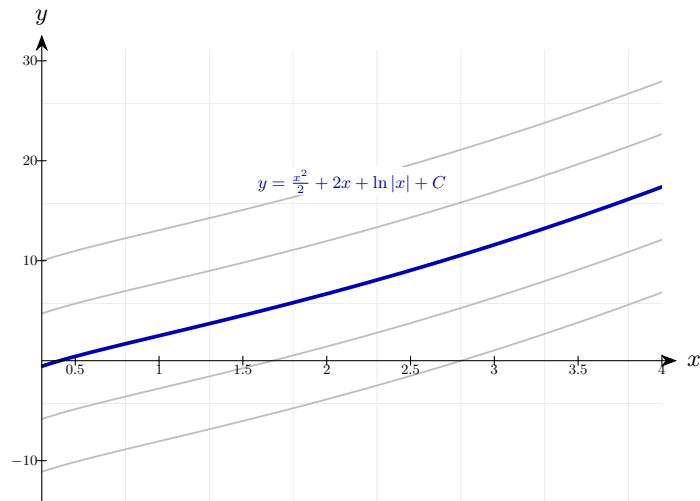
Step 2: Integrate term by term.

$$\int (x + 2 + x^{-1}) dx = \frac{x^2}{2} + 2x + \ln|x| + C$$

Apply the reverse power rule to x and the constant 2, giving $\frac{x^2}{2}$ and $2x$. The x^{-1} term is the special case $n = -1$ where the power rule fails (it would divide by zero), so we use the standard result $\int \frac{1}{x} dx = \ln|x| + C$ instead.

Final answer: $\frac{x^2}{2} + 2x + \ln|x| + C$

Diagram.



The family of antiderivative curves $y = F(x) + C$ — vertical translates differing by the constant C .

Indefinite Integration 68 (i1-068)

Difficulty: Standard / Marks: 5

A curve passes through $(4, 20)$ and has gradient $\frac{dy}{dx} = 3\sqrt{x}$. Find the equation of the curve.

Worked solution.

Step 1: Rewrite the surd and integrate.

$$y = \int 3x^{\frac{1}{2}} dx = 3 \cdot \frac{2}{3} x^{\frac{3}{2}} + C = 2x^{\frac{3}{2}} + C$$

Convert $\sqrt{x} = x^{1/2}$ so the power rule applies. Increase the index to $\frac{3}{2}$ and divide by $\frac{3}{2}$ (equivalently multiply by $\frac{2}{3}$); then $3 \times \frac{2}{3} = 2$.

Step 2: Substitute the point $(4, 20)$ to find C .

$$20 = 2(4)^{\frac{3}{2}} + C = 2 \cdot 8 + C = 16 + C \implies C = 4$$

Use the identity $4^{3/2} = (\sqrt{4})^3 = 2^3 = 8$. The boundary condition pins down which curve in the family passes through this point.

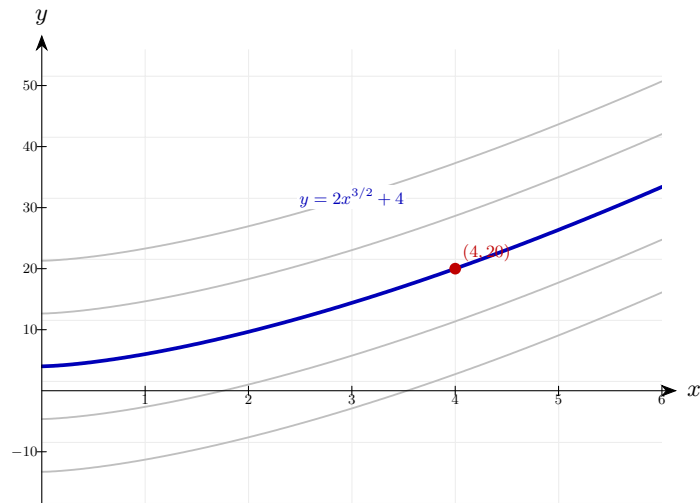
Step 3: State the equation.

$$y = 2x^{\frac{3}{2}} + 4$$

Substitute $C = 4$ into the general antiderivative.

Final answer: $y = 2x^{\frac{3}{2}} + 4$

Diagram.



The family of antiderivatives (grey) with the particular solution (blue) through the given point.

Indefinite Integration 69 (i1-069)

Difficulty: Standard / Marks: 5

Given $f'(x) = 2x - \frac{4}{x^3}$ and $f(1) = 3$, find $f(x)$.

Worked solution.

Step 1: Rewrite $\frac{4}{x^3}$ as $4x^{-3}$ and integrate.

$$f(x) = \int (2x - 4x^{-3}) dx = x^2 + \frac{2}{x^2} + C$$

For $2x$: $2 \cdot \frac{x^2}{2} = x^2$. For $-4x^{-3}$: new index -2 , divide by -2 , giving $-4 \cdot \frac{x^{-2}}{-2} = +\frac{2}{x^2}$ — the two minus signs cancel, so be careful with the sign.

Step 2: Substitute $f(1) = 3$ to find C .

$$3 = (1)^2 + \frac{2}{(1)^2} + C = 1 + 2 + C \implies C = 0$$

The boundary condition gives $C = 0$, so no constant term is needed in the final function.

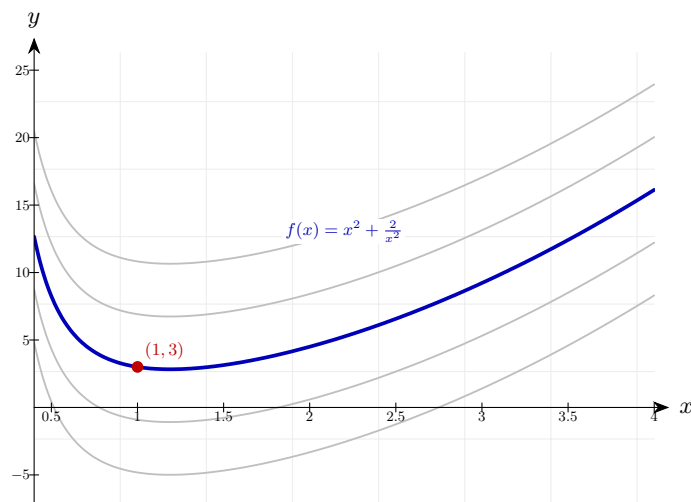
Step 3: State the final function.

$$f(x) = x^2 + \frac{2}{x^2}$$

Since $C = 0$, the antiderivative simplifies to this clean form.

Final answer: $f(x) = x^2 + \frac{2}{x^2}$

Diagram.



The family of antiderivatives (grey) with the particular solution (blue) through the given point.

Indefinite Integration 70 (i1-070)

Difficulty: Standard / Marks: 7

A curve has second derivative $\frac{d^2y}{dx^2} = 6x$. The curve has a stationary point at $(1, 2)$. (a) Find $\frac{dy}{dx}$. (b) Find y in terms of x .

Worked solution.

Step 1: (a) Integrate the second derivative to get the first derivative.

$$\frac{dy}{dx} = \int 6x \, dx = 3x^2 + A$$

Apply the reverse power rule: $6 \cdot \frac{x^2}{2} = 3x^2$. Use a different letter A for this constant of integration so it does not get confused with the constant from the second integration later.

Step 2: A stationary point means the gradient is zero, so use $\frac{dy}{dx} = 0$ at $x = 1$.

$$0 = 3(1)^2 + A \implies A = -3$$

This boundary condition pins down A , selecting the specific gradient function from the family of antiderivatives.

Step 3: State the gradient function.

$$\frac{dy}{dx} = 3x^2 - 3$$

Substitute $A = -3$ back into the general expression.

Step 4: (b) Integrate the gradient function to find y .

$$y = \int (3x^2 - 3) \, dx = x^3 - 3x + B$$

Apply the reverse power rule: $3 \cdot \frac{x^3}{3} = x^3$ and $\int -3 \, dx = -3x$. A new constant B appears — every integration introduces its own $+C$.

Step 5: Use the point $(1, 2)$ to find B .

$$2 = (1)^3 - 3(1) + B = 1 - 3 + B \implies B = 4$$

The boundary condition selects the unique curve from the family that passes through $(1, 2)$.

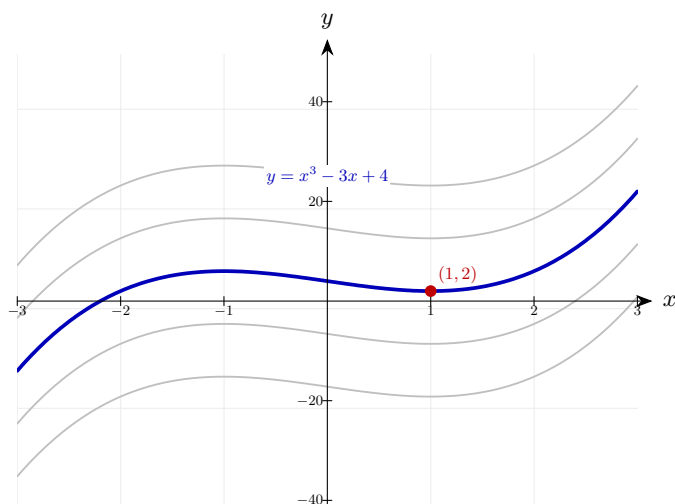
Step 6: State the equation of the curve.

$$y = x^3 - 3x + 4$$

Substitute $B = 4$ into the general antiderivative.

Final answer: (a) $\frac{dy}{dx} = 3x^2 - 3$ \ (b) $y = x^3 - 3x + 4$

Diagram.



The family of antiderivatives (grey) with the particular solution (blue) through the given point.
